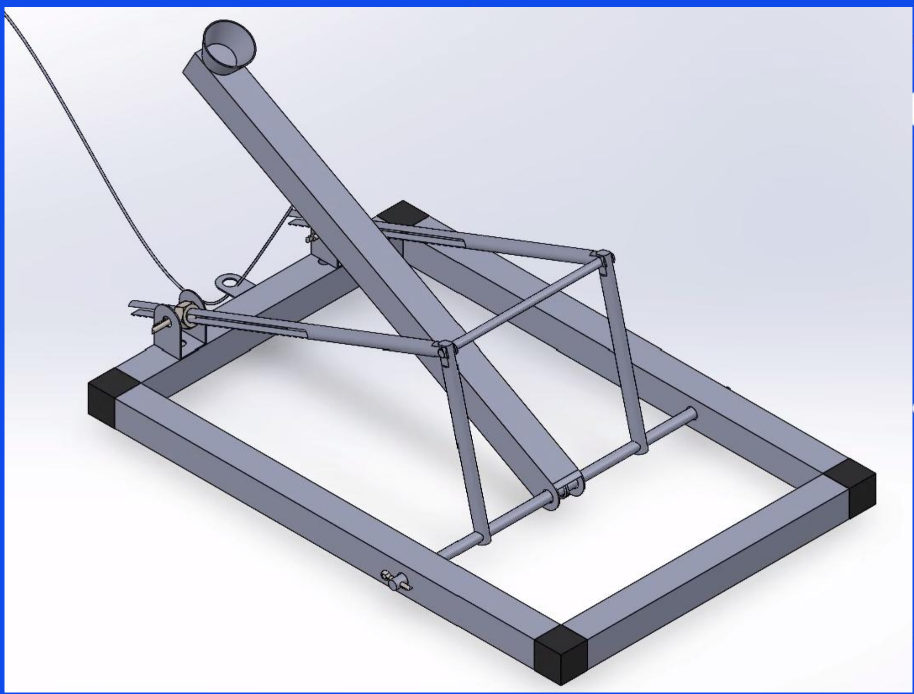
Engineering Prototype Project

Assignment 4

48610 Introduction to Mechanical Engineering



Arabella Simpson

Autumn 2026

Contents

1 Abstract

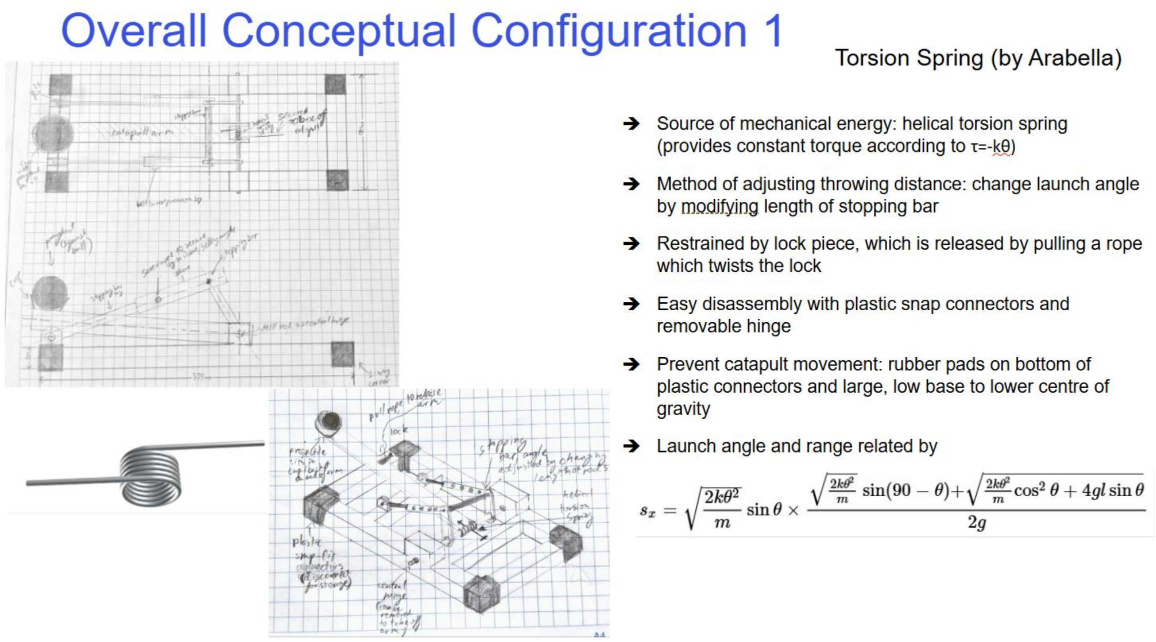
DeliverPult is a company which focuses on sustainable and ecofriendly goods delivery. For their mobile catapult delivery system, we have created a small-scale working prototype. This prototype is based on concepts developed and evaluated in Phase 1 and will allow the company to investigate the feasibility of the system before integrating Industry 4.0 features such as on-board robot loaders. Our catapult is a modular system centred around a mechanical torsion spring, designed to launch a 24g projectile into a 10cm target, precisely adjustable across a range of 3 to 4 metres.

In line with DeliverPult's requirements, the aluminium catapult is constrained to a single mechanical energy source, has a sub-20 second loading time, and can be completely disassembled to fit within defined storage dimensions. Range variability is achieved through an adjustable stopping-bar mechanism that dictates the final launch angle between 0 and 90 degrees.

Theoretical kinematic modelling and SolidWorks simulations were conducted to map the required stopping angles to the specified target distances. The structural frame consists of aluminium square tubing and snap connectors, ensuring rapid, tool-less assembly while maintaining a low centre of gravity for dynamic stability during operation. Additionally, a remote rope-pull lock mechanism was integrated to ensure safe, hands-free release. To accommodate workshop fabrication tolerances, an initially proposed infinitely adjustable threaded-rod stopping mechanism was modified into a discrete peg-hole system during manufacturing. The resulting prototype successfully satisfies all spatial, temporal, and kinematic constraints, delivering consistent projectile accuracy and modular reliability.

2 Detailed design process

The design of this catapult is based on our preferred conceptual configuration from Task 3, which was evaluated based on the scoring matrix seen in the Appendix.



This configuration centred around a torsion spring as the main source of mechanical energy, with the throwing distance being adjusted by modifying the angle of the stopping bar. From this concept, we refined the stopping bar and locking mechanisms, as well as simplifying the frame for manufacturing.

The torsion spring

The torsion spring was chosen for the concept out of the other mechanical energy sources for its simplicity and high power-to-weight ratio. Compared with other, linear, mechanical energy sources in the original morphological table such as gravity, elastic ropes or compression springs, the rotational nature of the helical torsion spring integrated simply with the pivoting catapult arm. This simplicity is also reflected in the calculations required, as the torque generated by the torsion spring is directly proportional to the angle through which the spring rotates. These linear options were also dismissed for their tendency to warp or decay over time, especially in cases such as the elastic rope.

It was difficult to source torsion springs at short notice and in low quantities, as is further elaborated on in the Appendix, but we managed to acquire a strong spring from the inside of a clamp. However, the internal diameter of this spring was too small to fit on any of the rods available in the workshop, as the design featured it threaded onto the central hinge inside the catapult arm, which necessitated a minimum diameter of 5 mm to avoid distortion due to bending stresses. As such, alternate torsion springs were found from clothes pegs, with lower spring coefficients but a much larger internal diameter. Based on online estimates of these spring coefficients (also seen in Appendix) and simplified calculations, it was believed that a single one of these springs should be sufficient to launch the squash ball within our 3-4 metre range.

However, the assumptions made in the theoretical stage did not reflect the reality, and poor tolerancing, friction and a large projectile holder, which had not been accounted for in the calculations, meant that all 3 of the clothes peg springs combined were not able to launch the squash ball more than 1.5 metres in physical testing. After some modification, we were able to use the original torsion spring, which proved perfectly suited for the project and reliably launched the squash ball 3.5 metres in testing.

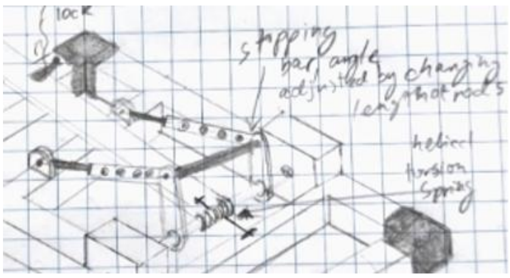
As is discussed in the locking mechanism section below, there are still some inefficiencies in the design, and significant force was required in our final tests to achieve a range of more than 3.5 metres. The torsion spring would therefore benefit from being upgraded, by adding another alongside it or by using a slightly stronger alternative, to ensure that the catapult would reliably reach the upper limits of the specified range. Alternatives might be found either in larger spring clamps or from a professional spring manufacturer, but calculations and testing have proved that it is a good source of mechanical energy with significant advantages for power-to-weight ratio and its consistency with minimal material wear.

The stopping bar mechanism

This is one of the key components of the design, and one which went through several variants as we compared the benefits of simplicity, structural strength and adjustability.

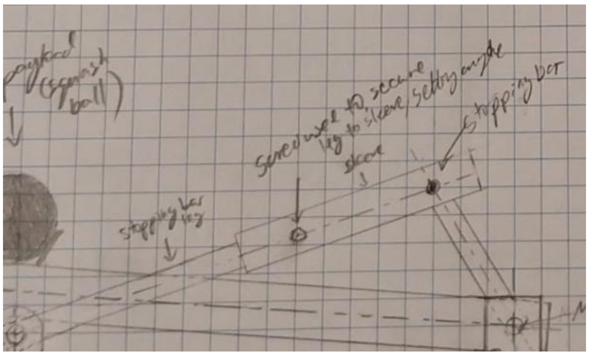
Many different methods of adjusting stopping bar angle were suggested including variants of telescopic design with different peg holes, screwing onto the inner rod at any position, a threaded rod and nut supporting the sleeve, a slit in threaded rod with nuts either side to hold onto hinge, etc.

Initial concept: peg holes



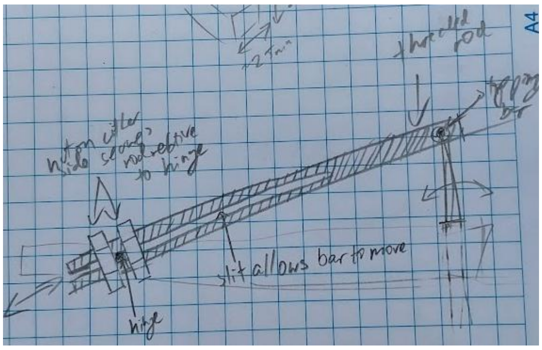
The peg hole design focused on simplicity, with a few designated hole positions to allow quick and consistent angle settings, and an easily manufacturable design. It consists of a central rod with one hole in it and a sleeve with several holes, through which a pin could be placed to set the angle. However, the limited number of positions requires complete consistency between the two sides, and do not allow the catapult's range to be finely adjusted.

Alternate design: grub screw



The grub screw design further simplified the sleeving mechanism of the previous concept, replacing the hole in the central rod with a grub screw which threaded through the outer sleeve and pressed on the rod to hold the two together. This design is infinitely adjustable, but the grub screw is inherently weaker than the pin through both parts, as it is only held in place by friction.

Preferred concept: threaded slot system



This system moved away from the rod and sleeve design in favour of a single threaded rod, which can be infinitely adjusted like the grub screw and is locked securely with a pin through the entire part, like the peg holes. Instead of adjusting the stopping-bar's angle from the centre of the member, the position of the base hinge is modified by moving the hinge pin up or down along the length of the bar in a slot, clamping it in place at the desired position between two hexagonal nuts. The main disadvantage of this system is in its strength and manufacturing, as the long slot weakens the rod and is difficult to construct, which is why, although this is the design shown in the technical drawings, it could not be created for the physical prototype within the scope of the project with the available tools.

Simplifying the base

The main feedback which we received from Phase 1 was on the necessity of simplifying the design of the catapult to improve strength and manufacturability. As can be seen in the scoring matrix (see Appendix), conceptual configurations 1 and 2 scored very similarly based on the weightings designated by DeliverPult, with the main advantage of concept 1 being its simplicity. We did this by integrating the central hinge into the base of the catapult instead of having it elevated, and by placing the back of the stopping-bar mechanism on the rear span of the catapult base in the place of another horizontal span, reducing the number of aluminium tubing pieces needed from 8 to 5.

Locking mechanism

The locking mechanism holds the arm in the loaded position and lets it be fired from 0.5m or more away. We used an L-shaped angle bracket riveted to the rear of the base frame, parallel to the catapult arm, with polyester string threaded through it. The string ties to the base of the PVC pipe connector and pulls the arm down against the base. The bracket originally sat in line with the top of the base until we decided to invert it, as this allowed us to pull the catapult arm down closer to the floor to achieve better compression and range. We also filed the end of the PVC pipe down slightly so the string could sit against it more securely.

To fire, the operator just pulls the free end of the string. In testing we found the catapult got its best range when the arm was manually pushed all the way down before latching - the string mechanism on its own wasn't quite achieving full spring pre-load. It takes quite a bit of force to tension the string this way. The locking mechanism was probably the least developed part of the design, and a pulley or longer lever arm would make a further difference. We discuss this further in Section 6.

Projectile holder

The projectile holder sits at the end of the catapult arm and carries the squash ball during launch. We originally planned on using a simple angled cup that would hold the ball without constraining it as it left the arm. During construction however, there was no way to make the short piece of straight PVC pipe we had sourced sit securely on the arm, so we ended up using a curved T-shaped PVC pipe connector bolted through the end of the arm, which held firmly and cradled the underside of the ball.

In early testing our range was well below what the calculations predicted. We traced the problem to the depth of the connector bore — the ball was getting stuck against the inner walls before it could exit, losing a lot of energy in the process. We fixed it by pressing a masking tape webbing across the inside of the connector, which raised the ball up to just below the opening and let it release cleanly. It worked, though it's not a permanent fix. Trimming the connector to the right depth, or 3D printing a purpose-made cup, would be more reliable — we cover this in Section 6.

Stabilising the catapult

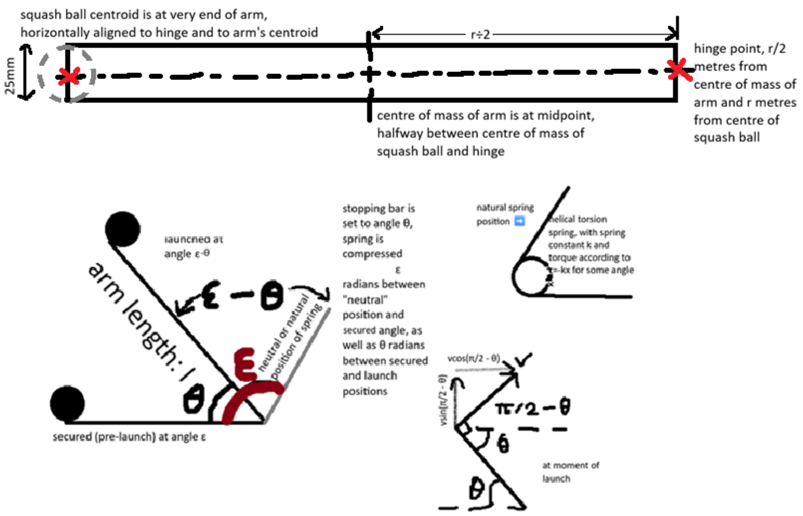
Originally, we planned on using rubber feet similarly to ones used in commercial furniture to increase the friction of the catapult base and prevent movement due to the translational forces generated in the catapult's launch. However, the available rubber furniture feet were thick and would raise the catapult's centre of gravity, making their use counterintuitive. Other alternatives such as Croc Grip anti-slip tape were also unsuitable due to the lack of friction provided by their surfaces. Therefore, we opted for a rubber tape, which gave us a wide surface area for gripping the floor, and was thin enough to keep the catapult base low to the ground. Weights were also used to prevent the base from shifting due to the large forward turning moment generated by the catapult's launch.

3 Relevant theories and equations

The conservation of energy theory is used to calculate the launch velocity of the projectile based on the total conversion of potential energy from the torsion spring into kinetic energy of the arm and squash ball. The following assumptions have been made:

- there is no friction or air resistance
- all potential energy will be converted to kinetic with no losses due to heat, sound or warping of parts
- there are no lateral forces and 2D free-body diagram can fully represent the situation
- the centre of mass of the catapult arm is at its geometric centroid, halfway up the length
- the arm itself is a perfectly symmetrical square-profile tube, which pivots around a hinge at one end and holds the middle of the squash ball at the other end
- the ball will leave its holder perpendicular to the arm
- the squash ball will travel according to projectile motion with only gravitational acceleration

With these simplifications, the following free-body diagrams have been made, showing the different states of the catapult arm, spring and squash ball, as well as the definitions for certain variables.



Based on the assumptions made and the variables defined, calculations were performed to model the catapult's range and assess the design, as can be seen in the following tables.

<table>
<tr>
<td>Calculation relating range and stopping bar/launch angle</td>
<td>

The basis of this series of calculations is the torsion spring, which generates torque (τ) proportional to the angle of its bending (x) according to some spring constant “k”, as seen in the equation

$$\tau = kx = I\alpha$$

The work done by the torsion spring in rotating the arm from ε to $\varepsilon - \theta$ is given by

$$W = \int_{\varepsilon}^{\varepsilon-\theta} \tau \, dx$$

This work is converted into kinetic energy such that $K = -W$.

$$\therefore K = \int_{\varepsilon-\theta}^{\varepsilon} kx \, dx$$

$$K = \frac{k}{2} \times [\theta^2]_{\varepsilon-\theta}^{\varepsilon} = \frac{k}{2}(\varepsilon^2 - (\varepsilon - \theta)^2)$$

Rotational kinetic energy is defined by $K = \frac{1}{2}I\omega^2$, so

$$\frac{k}{2}(\varepsilon^2 - (\varepsilon - \theta)^2) = \frac{1}{2}I\omega^2$$

$$\therefore \omega = \sqrt{\frac{k(2\varepsilon\theta - \theta^2)}{I}}$$

This angular velocity is converted into linear velocity at launch according to $v = \omega r$, giving the following launch velocity perpendicular to the arm:

$$v_0 = \sqrt{\frac{k(2\varepsilon\theta - \theta^2)}{I}} \times r$$

The horizontal component of the initial velocity (u_x) is equal to $v_0 \cos(\frac{\pi}{2} - \theta)$, while the vertical component is given by $u_y = v_0 \sin(\frac{\pi}{2} - \theta)$.

Using basic projectile motion equations based on gravitational acceleration with no air resistance, the horizontal range of the projectile is defined as $s_x = u_x t$ while the vertical position as a function of time is $s_y = y_0 + u_y t + \frac{1}{2}at^2$.

The length of time that the projectile will spend in the air, assuming that it is launched from the height of the stopping bar $(y_0 = r \sin\theta)$ and its acceleration is a constant “-g”, is $t = \frac{u_y \pm \sqrt{u_y^2 + 2grsin\theta}}{g}$, and the positive branch is taken to find time after launch rather than, nonsensically, before.

Substitute this time into the horizontal range equation to get the range of the catapult based on the angle of launch.

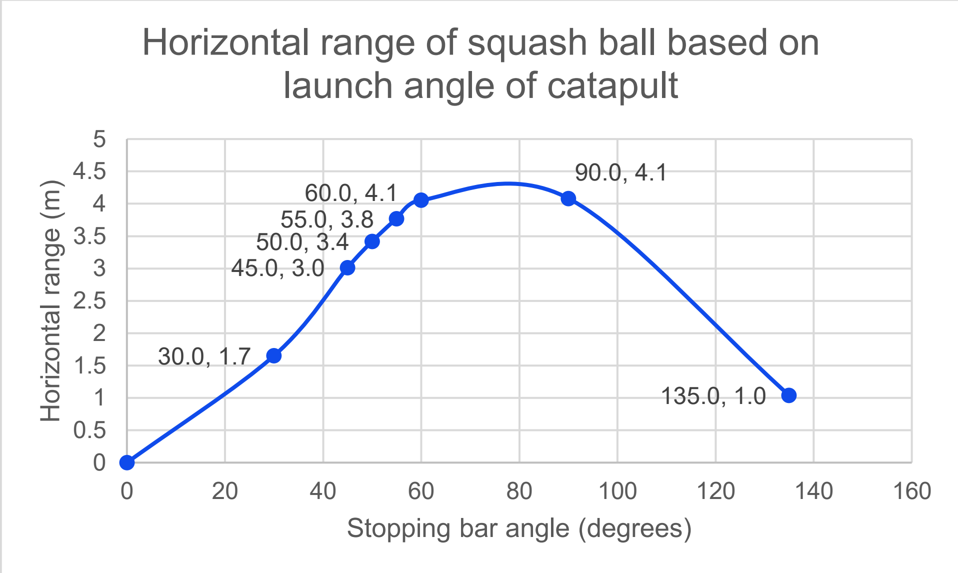
$$s_x = v_0 \cos\left(\frac{\pi}{2} - \theta\right) \times \frac{u_y + \sqrt{u_y^2 + 2grsin\theta}}{g}$$

$$\therefore s_x = \sqrt{\frac{k(2\varepsilon\theta - \theta^2)}{I}} \times r\cos\left(\frac{\pi}{2} - \theta\right) \times \frac{\sqrt{\frac{k(2\varepsilon\theta - \theta^2)}{I}} \times r\sin(\frac{\pi}{2} - \theta) + \sqrt{\frac{kr^2(2\varepsilon\theta - \theta^2)}{I} + 2gr\sin\theta}}{g}$$

The following variables can be inserted into the equation to estimate the range of angles used to launch the squash ball between 3 and 4 metres.

- k = spring constant = 0.34 N.m/radian from online calculator estimate (see appendix)
- ε = angle between spring resting position and fully extended≈130° or 2.27 radians
- r = length of catapult arm = 0.4 m
- g = gravitational constant = 9.81 m/s²
- I = mass moment of inertia of catapult arm = 6.9801×10^{-3} kg.m² (see calculations below)

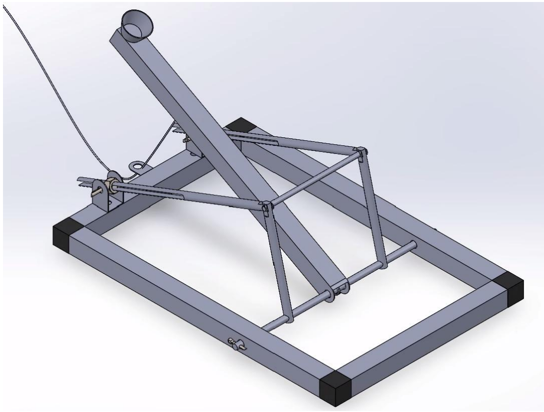
</td>
</tr>
</table>

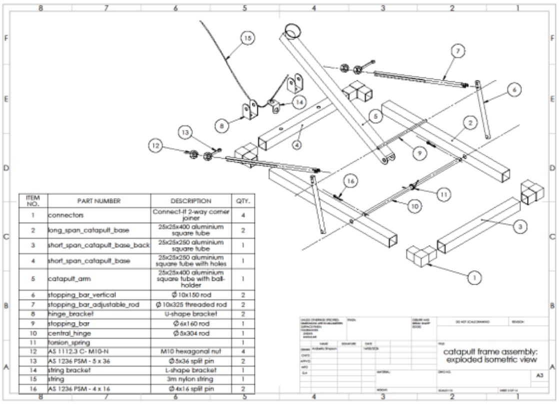
	
Calculation mass moment of inertia for the catapult arm	"I" is the mass moment of inertia of the rotating object, in this case the combined mass moments of the arm of the catapult and the projectile (squash ball). Catapult arm: square-section aluminium tube 25mm x 25mm x 0.4m with 1.2mm thickness $\rho = \frac{m}{V} = 2700$ kg/m³ The volume of tube is found by subtracting the volume of the hole in it from the solid volume $V_{tube} = V_{block} - V_{hole} = 0.025^2 \times 0.4 - ((0.025 - 0.0012)^2 \times 0.4) = 23.424 \times 10^{-6}$ m³ $m_{tube} = 2700 \times (23.424 \times 10^{-6}) = 0.063245$ kg The mass moment of inertia of the arm around its centroid "O" (0.2m from the pivot point) is given by $I_O = \frac{1}{12}m(w^2 + h^2)$ where w=25mm and h=0.4m $I_O = \frac{1}{12} \times 0.063245 \times (0.025^2 + 0.4^2) \approx 0.0008466$ kg.m² Using the parallel axis formula ($I = I_G + md^2$), the moment of inertia of the arm around the pivot point, which is assumed to be exactly at the end of the arm, can be found $I_{arm} = I_O + 0.063245 \times \left(\frac{0.4}{2}\right)^2 = 0.0033764$ kg.m² Similarly for the squash ball, which as a sphere has moment of inertia formula $I = \frac{2}{5}mr^2$, the parallel axis formula is used to give $I_{ball} = \frac{2}{5} \times 0.0225 \times \left(\frac{0.0405}{2}\right)^2 + 0.0225 \times 0.4^2 = 0.0036037$ kg.m² Adding the arm and ball's results together, the catapult arm's moment of inertia is $I = 0.0033764 + 0.0036037 = 0.0069801 = 6.9801 \times 10^{-3}$ kg.m²
Shear stress on hinge pin (part 8 in technical drawing)	$\sigma = \frac{F}{A}$ Force will be the force exerted by the spring through torque at the moment of impact with the stopping bar, which is transferred down the supporting members to the hinge pin. The shear area is the circular cross-section of the pin, and the force will be divided by 2 as the stopping bar is supported on either side. From the $\tau = rF$ definition of torque, the torsion spring's force is given by: $F(x) = \frac{kx}{r}$ x = angle spring has been bent from rest position = ε-θ as seen in FBD, so the maximum shear force would be at $x = \varepsilon$ or x = 2.27 radians r = 0.4m k=spring constant=0.34 N.m/radian (from online calculator estimate) A = cross-sectional area experiencing shear = πr^2

	r ≈ 2 mm (radius of split pin) From these variables, $\sigma = \frac{0.34*2.27}{0.4} * \frac{1}{\pi*(2*10^{-3})^2} \approx 154kPa$ Based on research into common shear strengths of materials (see Appendix), even the softest metals have shear strengths of at least 60 MPa, which means that this stress load, which does not exceed 1 MPa even when halving the radius or doubling the spring constant, is well within the working range of any material used, and the design for the catapult and stopping-bar mechanism can be safely and reliably used.

4 Final design

Overview





ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	connectors	Connect-it 2-way corner joiner	4
2	long_span_catapult_base	25x25x400 aluminium square tube	2
3	short_span_catapult_base_back	25x25x250 aluminium square tube	1
4	short_span_catapult_base	25x25x250 aluminium square tube with holes	1
5	catapult_arm	25x25x400 aluminium square tube with ball-holder	1
6	stopping_bar_vertical	Ø 10x150 rod	2
7	stopping_bar_adjustable_rod	Ø 10x325 threaded rod	2
8	hinge_bracket	U-shape bracket	2
9	stopping_bar	Ø 6x160 rod	1
10	central_hinge	Ø 5x304 rod	1
11	torsion_spring		1
12	AS 1112.3 C- M10-N	M10 hexagonal nut	4
13	AS 1236 PSM - 5 x 36	Ø 5x36 split pin	2
14	string bracket	L-shape bracket	1
15	string	3m nylon string	1
16	AS 1236 PSM - 4 x 16	Ø 4x16 split pin	2

The final design of the catapult is based on a torsion spring, designed to launch a squash ball 3 – 4 metres with repeatable accuracy. The catapult's maximum dimensions are 450 mm x 300 mm x 400 mm, built from 25 mm x 25 mm aluminium square tubing, with plastic snap connectors and a removable hinge-rod for disassembly. When the catapult's arm is released from a constant position attached to the base, it rotates around a central hinge due to the work done by the torsion spring, accelerating the payload until it hits a stopping bar. This stopping bar is set to a specific angle according to the desired range of the payload, which launches perpendicularly out of the holder on the arm and lands according to this formula:

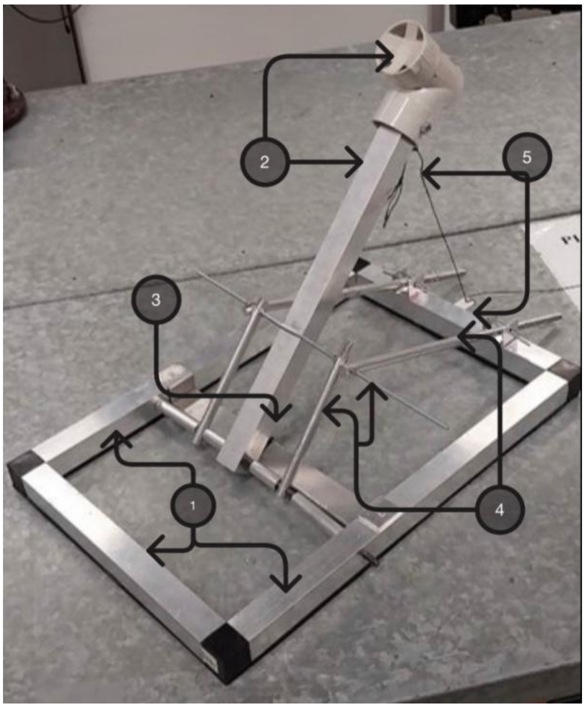
$$s_x = \sqrt{\frac{k(2\varepsilon\theta - \theta^2)}{I}} \times r\cos\left(\frac{\pi}{2} - \theta\right) \times \frac{\sqrt{\frac{k(2\varepsilon\theta-\theta^2)}{I}} \times r\sin(\frac{\pi}{2} - \theta) + \sqrt{\frac{kr^2(2\varepsilon\theta-\theta^2)}{I} + 2gr\sin\theta}}{g}$$

Original alternatives to provide mechanical energy were an elastic rope, a compression spring, pendulum striker and a ramp. However respective complications due to material wear, difficult produced consistency of power level, adjustable power and complex design made the torsion spring the better alternative, as is discussed in the design section of this report.

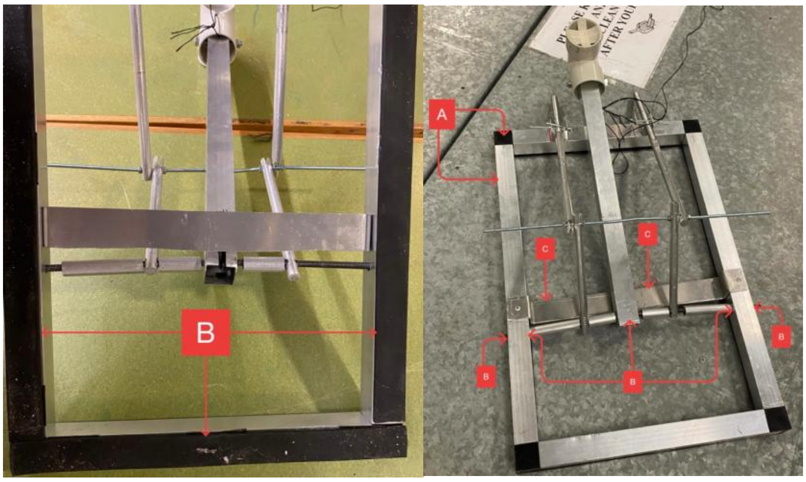
Sub-assemblies

The following sub-assemblies, numbered from 1-5, make up the catapult.

1. Base frame
2. Catapult arm
3. Helical torsion spring
4. Stopping bar
5. Release mechanism

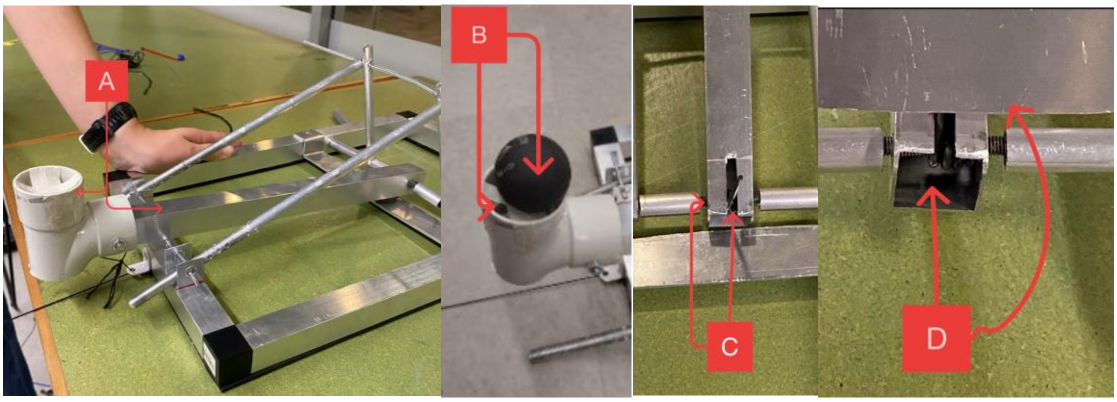


1. **Base Frame**



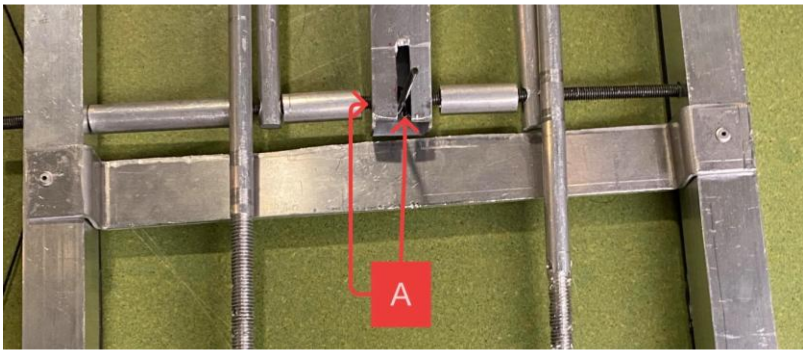
The base frame of the catapult is manufactured from 25x25 mm aluminium square tubing, which is connected with plastic snap connectors to form the 450 x 300mm frame (A). To connect the torsion spring, ⌀5mm holes were drilled 300mm from the rear and a ⌀ 5 x 304mm hinge rod is inserted to secure the catapults arm/stopping bars to the base frame (B). Then, a flat aluminium sheet is attached adjacent to the hinge rod holes towards the back, providing a surface for the torsion spring to repel off during launch (C). Finally, the underside of the aluminium is lined with rubber, which is attached with a combination of double-sided tape and hot glue, to prevent the base from shifting when the catapult launches (D).

2. **Catapult arm**



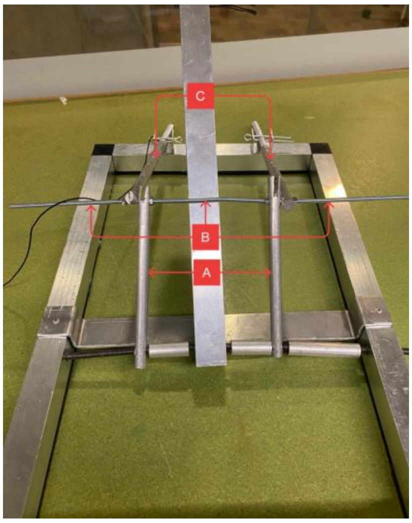
The catapult arm is made from a 400mm section of the 25x25mm aluminium tubing with a PVC pipe connector bolted to one end (A). The pipe connector acts as a holder to carry the squash ball payload and was modified to the appropriate depth with a masking tape net, as discussed in the design section (B). On the other end, a ⌀5mm hole allows the arm to pivot freely on the hinge rod, and the torsion spring also sits on this central hinge, within the catapult arm (C). One end of the torsion spring pushes on the inner wall of the catapult arm, while the other presses against the floor through a 35x6 mm slit in the base of the arm (D).

3. Helical torsion spring



The helical torsion spring is the catapult's source of mechanical energy. Removed from a Cartright 100mm spring clamp, the spring is fitted into the catapult arm and attached through the central hinge rod. The spring generates its torque by pushing against both the interior aluminium bar and the flat base sheet (A).

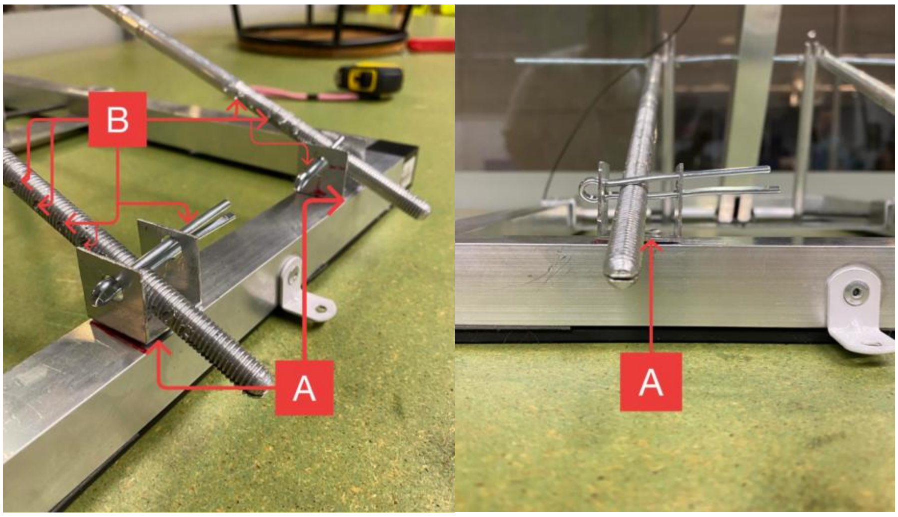
4. Stopping bar



Front:

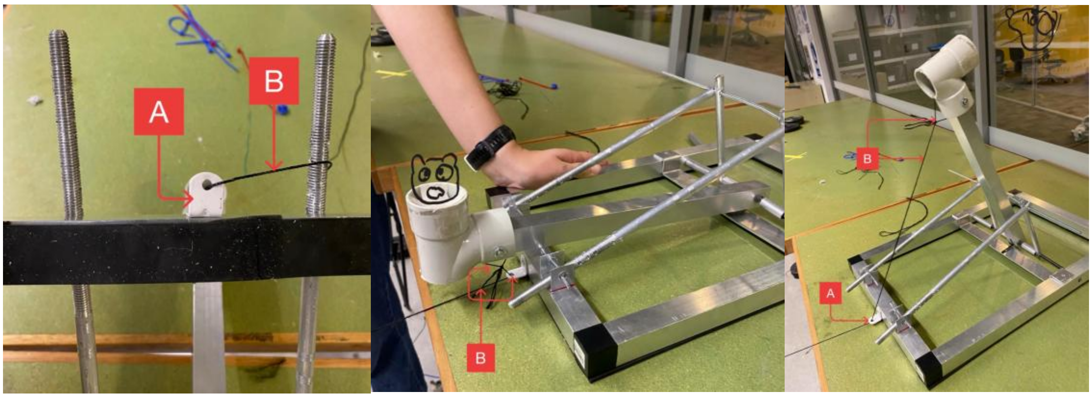
The main element of the stopping mechanism is the horizontal stopping bar (B), a ⌀6x160mm threaded rod which is suspended at an adjustable angle above the base, where it collides with the same spot on the catapult arm every time. The stopping bar is held in position by a vertical and diagonal rod on either side of the arm. At the front of the stopping mechanism are two 150mm vertical rods, each with ⌀ 7mm holes 12mm down from the ends of the rods and positioned equidistant from the width of the base frame with metal spacers (A). These vertical rods pivot around the central hinge according to the angle of the adjustable spans. Two ⌀10x325mm threaded rods act as the adjustable diagonal members for the stopping bar, and they are attached to the stopping bar itself with a ⌀7mm hole drilled on the front end of the rod (C).

Rear:



The back of the stopping mechanism has two U-shaped hinge brackets parallel to the vertical stopping bars attached to the base frame of the catapult (A). Then the rear ends of the adjustable stopping bar members have holes drilled every 10mm along them (B) which allow split pins to be slotted through to secure to the U-shaped brackets. The split pins can be slotted through any hole in the member to define the working length of the diagonals, modifying the launch angle of the catapult arm.

5. Release Mechanism



The release mechanism is made from a L-shaped angle bracket which is attached to the rear of the catapult's base frame parallel to the arm (A)**.** A string is then tied to the bottom of the PVC pipe on the catapults arm and threaded through the L-shaped angle bracket, which is then pulled and released, launching the squash ball (B).

5 Requirement testing

We checked the prototype against the requirement list in the Appendix using a mix of design checks, measurements and live launch testing, as shown in the table below. Some requirements we could confirm just from the geometry or assembly; others needed the physical prototype to test properly. We didn't have time to fully test the throwing range (Requirement 2.1) or catapult stability (Requirement 3.2) before submission, but some testing has been performed as per the methods outlined below. More testing should be performed on the range of the catapult at different angles to compare theoretical calculations with actual performance, but overall the prototype was successful and aligned with theory and requirements.

No.	Requirement	Testing	Result	Successful?
1.5 – 1.7	Disassembled dimensions, must fit in tub	Concept: must be smaller than storage tub dimensions Physical: Testing in actual tub	Catapult fits when arm is removed by taking out central hinge.	Yes
1.4	Fits squash ball payload	Design: holder is larger than squash ball diameter Physical: Test launches to confirm projectile stays stable	Squash ball fits snugly in PVC pipe connector, which is the ideal diameter to hold the ball stably, but the ball had to be elevated with a masking tape net to cleanly exit the arm.	Yes

2.1	Throwing range, can hit targets between 3 and 4 metres	Calculations: projected range can reach at least 4 metres Physical: testing maximum range and specific distances	The calculated range for the catapult was up to 4.1 metres. In testing, we achieved a maximum range of 3.8 metres and could reliably hit any point up to this range within 10 cm. However, we were not able to reach 3.9 metres or more, as seen in test results table below.	Partially
3.2	Stability	Physical: catapult base should not shift more than 3 cm in any direction or audibly or visibly lift off of the floor.	When stabilised with rubber tape, the base shifted up to 2 cm during firing, but it did still rock up at the back until it was weighed down with 2kg on each corner.	Partially
5.2	Reliable locking mechanism, no accidental firing	Physical: catapult left in loaded position for 5 minutes and observed throughout	When tied down properly, the string was able to hold the tension of the catapult arm for over 5 minutes with no slippage or distortion of the string bracket.	Yes
6.2	Assembly time, less than 5 minutes for one person	Physical: Every team member attempts to assemble catapult individually while being timed by stopwatch	There were some issues with jamming between the central hinge and parts such as the torsion spring and stopping bar vertical members, but all individuals were able to insert the central hinge and catapult arm onto the base in less than 5 minutes.	Yes
7.2	Loading time, less than 20 seconds	Physical: Timing with stopwatch	The loading time was substantially under 20 seconds as the ball was placed quickly into the holder and the string pulled back.	Yes

Results of some testing of the catapult's range at 2 different angles (highlighted tests successfully achieved the 3-4 metre range):

Angle	35	40
Test 1	187	248
Test 2	238	**380**
Test 3	**339**	**364**
Test 4	**325**	**371**
Test 5	257	**387**

This empirical testing demonstrates that the catapult is capable of launching the projectile within the desired 3-4 metre range, but also that significant variability can also be achieved by modifying conditions other than the angle, such as the initial compression of the arm and the manner of release, whether manual, rapidly with the string or with some resistance from the string.

6 Discussion and Conclusion

Overall, the catapult prototype was a successful proof of concept for the simplicity and power of the torsion spring as a source of mechanical energy. We were able to safely and reliably launch the projectile over 3 metres, meeting most tested requirements. Of course, as a prototype, there is still plenty of room for improvement in the design and manufacturing before DeliverPult can upgrade it to a full-sized model. Some of these possible improvements include:

Refining the stopping bar diagonals

We were unable to cleanly manufacture the slot in the threaded rod for the infinitely adjustable stopping bar mechanism as designed in our CAD, instead opting for the incremental position holes suggested in previous designs. Investigating a way to implement this design, such as with a bandsaw or CNC to accurately create the part, would be beneficial to meet DeliverPult's requirements for accuracy and reduce inconsistency due to error between the two sides of the stopping bar.

Improving the locking mechanism

The locking mechanism was probably the weakest part of the design. As discussed in Section 2, the string-and-bracket setup didn't give us enough mechanical advantage to consistently reach full spring pre-load, which affected both range and repeatability. Adding a pulley or a longer lever arm would make a real difference here. A cam-and-latch design, where the arm clicks into a catch and is released by pulling a cord, would also give a more reliable lock than a string.

Redesigning the projectile holder

For the projectile holder, trimming the PVC connector to the right depth would be the simplest fix. We could also 3D print a purpose-made cup sized to the ball and shaped to sit on the arm properly, which would give us better control over the release angle. Either way, the masking tape net solution works for now but isn't something we'd want to rely on permanently.

More accurate theoretical modelling

The theoretical model significantly overestimated our range in early testing. It assumed no friction and full conversion of spring energy to kinetic energy, but in practice we lost energy to hinge friction, the PVC connector had more rotational inertia than we'd accounted for, and we weren't consistently reaching full spring pre-load. The angle-vs-distance results we measured in testing are a more reliable guide for setting the stopping bar than the theoretical values. Adding a friction term to the model would probably bring the predicted range much closer to what we measured.

7 References

Bunnings Warehouse. (n.d.). *Results for "torsion spring".* https://www.bunnings.com.au/search/products?page=1&q=torsion+spring&sort=BoostOrder

Jubilee Springs. (n.d.). *Torsion Springs.* https://www.jubilee-springs.com.au/torsion-springs/

Lesjofors Heavy Springs. (n.d.). *Heavy Duty Torsion Spring Manufacturers.* https://lesjoforssprings.com/torsion-springs/

Planet Spring. (n.d). *Spring Creator 4.0 - Torsion Spring Calculator.* Retrieved April 26, 2026, from https://www.planetspring.com/pages/torsion-spring-calculator-torsion-spring-calculation.php

Torsion spring. (2026, March 18). In *Wikipedia*. https://en.wikipedia.org/wiki/Torsion_spring

UniPunch. (2026, May 1). *Material Technical Information. SPECIFICATION FOR COMMON MATERIALS AND RECOMMENDED DIE CLEARANCE.* https://unipunch.com/wp-content/uploads/2023/05/Material-Tech-Specs.pdf

8 Appendix

List of individual contributions to project and report

Student	Contribution to project	Contribution to report

Arabella Simpson	- Morphological table
	- Gantt chart
	- Task and Requirement Analysis
	- Designed original concept solution on which final design is based
	- Concept sketches to refine design of sub-assemblies
	- Researched torsion spring physics
	- All SolidWorks:
	  o Part designs
	  o Assembly
	  o Technical drawings
	- Delegating tasks/organising group work over the course of the project, including encouraging everyone to contribute to areas which were lacking
	- Made WhatsApp group and shared documents to enable collaboration
	- Assembled frame
	- Found springs and holder for squash ball
	- Made excel spreadsheet to calculate theoretical range based on angle
	- Final testing
		- Made document
		- Added appendices/details from task 3
		- Free-body diagrams
		- Calculations
		  o Shear forces
		  o Range formula
		  o Area moment of Inertia
		- Research summary
		- Description of calculations and design
		- Added sketches/images/graphs
		- Reference list
		- Requirement testing table
		- Finalised abstract with contextual details
		- Editing team members' sections for professionalism, grammar, clarity and consistent aesthetics
		- Bill of Materials

Teammate 1	- Created the morphological table and conducted the initial pro/con analysis for each sub-function alongside Arabella - Developed one overall concept configuration with different choices from	- Documented the alternative conceptual solutions and drafted reasoning for the final design choice - Wrote and formatted the Morphological Table for the report's appendix

	each sub-function to present to the team - Developed the scoring matrix to evaluate the conceptual configurations - Collaborated to determine the final selected catapult design used for Assignment 3 - Participated with the entire group in assembling the physical catapult prototype - Assisted the team in testing and refining the physical catapult - Made a Solid works prototype that inspired later designs	- Wrote and formatted the Scoring Matrix for the report's appendix
Teammate 2	- Participated with the entire group in assembling the physical catapult prototype - Assisted the team in testing and refining the physical catapult - Assisted the team in testing and refining the physical catapult - Participated in group testing the catapult	1. The Final design A) Added images to support final design description B) Added image letter annotations C) Wrote descriptions for each sub assembly to the catapult D) Drafted the overall final design paragraph 2. Design stages A) Added the Angle Vs distance table B) Recorded the results of angle 40
Teammate 3	- Developed one overall concept configuration to present to the team - Assisted in evaluating concepts within the scoring matrix - Participated in team evaluation to determine Selected catapult configuration used - Partnered with Teammate 4 to measure, cut and present final non-assembled parts of the catapult to the team - Participated with the entire group in assembling	- Documented a part of the budgeting of the parts and materials needed for the catapult

	the physical catapult prototype - Assisted the team in testing and refining physical catapult	
Teammate 4	- Developed one overall concept configuration to present to the team - Assisted in evaluating concepts within the scoring matrix - Participated in team evaluation to determine Selected catapult configuration used - Partnered with Teammate 3 to measure, cut and present final non-assembled parts of the catapult to the team - Participated with the entire group in assembling the physical catapult prototype - Assisted the team in testing and refining physical catapult	- Documented part of the detailed design process for the locking mechanisms and projectile holder - Drafted parts of the final discussion paragraphs - Drafted the final conclusion paragraph - Editing and clarifying other sections of the report

Bill of Materials

This BOM displays the maximum cost of raw materials to manufacture the catapult, but significant cost saving can be achieved by producing multiple catapults, as many parts could only be purchased in bulk or may be already available in the workshop (i.e. rivets).

No.	**Part Name**	**Description**	**Source**	**Quantity**	**Cost (AUD)**	
					Unit	Total
1	Connector	2-way plastic corner joiner	Bunnings: https://www.bunnings.com.au/metal-mate-2-way-connect-it-corner-joiner_p1138466	4	2.41	9.64
2	Catapult base spans and arm (part no. 2, 3, 4 & 15 from assembly)	25.4 x 25.4 x 1.2mm aluminium tubing, 1.7 metres total (3x400mm spans, 2x250mm spans)	Bunnings (1m sections): https://www.bunnings.com.au/metal-mate-25-4-x-25-4-x-1-2mm-3m-silver-aluminium-square-tube_p1079488	2	18.09	36.18
5	Stopping bar vertical/ stopping bar adjustable rod	2 Ø10x150 aluminium rods 2 Ø10x325 aluminium rods (partially threaded) Ø10x950 total	Bunnings (1m section): https://www.bunnings.com.au/metal-mate-10mm-1m-aluminium-round-solid_p0427766	1	11.43	11.43

6	Hinge bracket	U-shaped aluminium or steel bracket with holes for split pin and for riveting to base	2 options: - Bunnings (sides removed): https://www.bunnings.com.au/australian-handyman-supplies-38-x-25mm-rail-saddle-bracket_p0910519 - Use excess from aluminium tubing, remove one side to create u-shape	2	2.40	4.80
7	Stopping bar	⌀6x160mm threaded steel rod	Bunnings (1.2m section): https://www.bunnings.com.au/macsim-m6-x-1-2m-hot-dipped-galvanised-threaded-rod_p2410297	1	5.07	10.14
8	Central hinge	⌀8x320 aluminium or steel rod	Mitre 10 (1m section): https://www.strathalbynmitre10.com.au/shop/building-timber/steel/reinforcing-round-rods/steel-rod-round-8mm/	1	4.45	4.45
9	Torsion spring	Helical torsion spring, inner diameter at least 10mm, made from at least ⌀2 mm wire	Removed from spring clamp (Bunnings): https://www.bunnings.com.au/craftright-100mm-metal-spring-clamp_p0916108	1	2.98	2.98
10	Hexagonal nut	M10 hexagonal nut	Bunnings: https://www.bunnings.com.au/zenith-m10-hot-dip-galvanised-hex-nuts-each_p2445379	4	0.33	1.32
11	Split pin	⌀4x40mm split pins, 4 needed	Bunnings (15 pack): https://www.bunnings.com.au/zenith-4-0-x-40mm-zinc-plated-split-pin-15-pack_p2420302	1	4.95	4.95
12	String bracket	L-shaped angle bracket	Bunnings (12 pack): https://www.bunnings.com.au/carinya-20-x-20-x-15-x-1-4mm-white-reinforcing-angle-bracket-12-pack_p3962549	1	3.51	3.51
13	String	3m of polyester string	Bunnings (80m roll): https://www.bunnings.com.au/grunt-80m-universal-polyester-line_p4310941	1	3.10	3.10
14	Pipe bend	T-shaped PVC pipe connector modified to sit on end of catapult arm	Bunnings: https://www.bunnings.com	1	2.90	2.90

			.au/holman-20mm-pvc-plain-tee_p3142334			
15	Rivets	3.2x3.2mm blind rivets	Bunnings (25 pack): https://www.bunnings.com.au/otter-3-2-x-3-2mm-open-aluminium-blind-rivets-25-pack_p0012068	6	3.76	3.76
16	Rubber stabiliser	4.5 mm x 25 mm x 3 m rubber tape	Bunnings (3 Meters): https://www.bunnings.com.au/moroday-4-5mm-x-25mm-x-3m-rubber-rectangle-black_p4000064	1	21.78	21.78
17	Gorilla tape	1.5m double-sided tape, used to affix rubber to base of catapult	Bunnings (1.5 meters): https://www.bunnings.com.au/gorilla-1-5m-black-double-sided-mounting-tape_p0054522	1	14.96	14.96
Total cost:						135.90

Requirement List

No.	Description / Requirement	Value			Unit	Who	Source / comment	Testing
		Min	Exact	Max				
1	Geometry							
1.1	Length	-	-	4.55	m	Arabella	Task Description (max length of test site is 4.5 ± 0.05 m), EPP page 5	CAD: Measured using SolidWorks tools Physical: using a ruler or measuring tape
1.2	Height	-	-	3.05	m	Arabella	Task Description (± 0.05 m), EPP page 5	CAD: SolidWorks measuring tools Physical: ruler or measuring tape
1.3	Width	-	-	1.05	m	Arabella	Task Description (± 0.05 m), EPP page 5	CAD: SolidWorks measuring tools Physical: ruler or measuring tape
1.4	Payload diameter	-	-	40.5	mm	Arabella	Task Description (squash ball), EPP page 6	N/A
1.5	Disassembled length	-	-	502	mm	Arabella	Task Description (to fit in storage container), EPP page 14	Physical: check that all parts fit in storage container
1.6	Disassembled height	-	-	300	mm	Arabella	Task Description (storage container), EPP page 14	Physical: check that no parts poke out of top of storage container
1.7	Disassembled width	-	-	338	mm	Arabella	Task Description (to fit in storage container), EPP page 14	Physical: check that all parts fit in storage container
2	Kinematics							

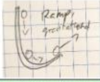
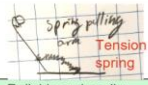
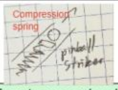
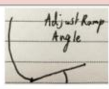
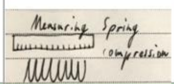
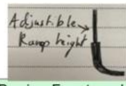
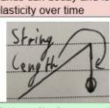
2.1	Throwing distance	3	-	4	m	Arabella	Task Description, EPP page 4	Concept: flight trajectory calculations Phys. prototype: series of throwing experiments using different distance settings
2.2	Launch Angle	30	-	60	degrees	Teammate 4	Calculated from projectile motion range for 3-4m range	Concept: Angle measurement with protractor
2.3	Launch Velocity	5.5	-	9	m/s	Teammate 4	Derived from projectile motion range of 3-4m range at 30-60° requires $v_0 \approx 5.5\text{-}9$ m/s	Concept: Calculated from range and angle measurements
3	**Forces**							
3.1	Weight of Payload	20	-	25	g	Arabella	Task Description (standard squash ball), EPP page 5	N/A
3.2	Stability	-	yes	-	-	Arabella	Task Description (should remain substantially in contact with ground at all times), EPP page 5	Concept: identify and minimise turning moment Physical: ensure catapult does not lift off ground during test launches
4	**Energy**							
4.1	Energy type	-	yes	-	-	Arabella	Task Description (only mechanical components shall be used), EPP page 5	N/A
4.2	Number of mechanical energy sources	-	1	-	-	Arabella	Task Description, EPP page 5	N/A
5	**Safety**							
5.1	Remote release feature	-	yes	-	-	Arabella	Task Description, safely released without anybody in the testing zone, EPP page 5	Phys. prototype: test launches on and before official testing date
5.2	Prevent accidental firing	-	yes	-	-	Arabella	Task Description, should be able to be kept safely in loaded state, EPP page 5	Concept: Mechanical locking pin/latch Physical: Load the prototype and verify it does not fire without the remote release being activated
5.3	No exposed sharp edges or protrusions	-	yes	-	-	Teammate 4	Safety; No exposed sharp metal edges that could cause injury, EPP page 5	Physical: Conduct a visual inspection of aluminium edges and joints before operation to ensure they are proper
6	**Assembly**							
6.1	Modular	-	yes	-	-	Arabella	Task Description, design needs to be modular, EPP page 5	Concept: Model is structured into distinct sub-assemblies Physical: Verify the disassembled parts fit completely into the 300mm storage tub dimensions
6.2	Assembly time of Arm	-	-	5	min	Teammate 3	Task Description, Catapult needs to be in an operation-ready state within 5 mins, EPP page 5	Physical: Use a stopwatch to time a single team member attaching the arm to the frame. Must be under 5 minutes
~~6.3~~	~~Loading time~~	-	-	~~5~~	~~s~~	~~Arabella~~	~~Task Description, quick and easy loading of catapult (<20 secs), EPP page 5~~	~~Physical: Use a stopwatch to time the sequence of loading the ball and pulling back the arm which has to be under 20 seconds~~
6.4	Single Person Assembly	-	yes	-	-	Teammate 4	Practical Requirement: One person must be able to set up without assistance	Phys. prototype: Every member of group attempts to assemble independently within specified timeframe (see 6.2)
7	**Operation**							

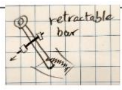
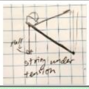
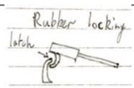
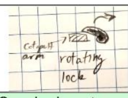
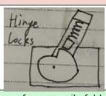
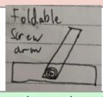
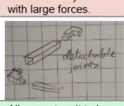
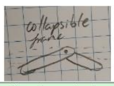
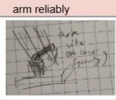
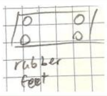
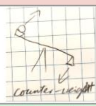
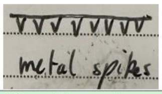
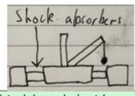
7.1	Adjustable Range	-	yes	-	-	Arabella	Task description, should be flexibly adjustable to different distances within range (see 2.1)	Concept: Demonstrates an adjustable mechanism Physical: Conduct test fires at different distances to verify the adjustability
7.2	Loading Time	-	-	20	s	Arabella	Task description, quick and easy loading, EPP page 5	Physical: Use a stopwatch to time the sequence of loading the ball and pulling back the arm which has to be under 20 seconds

Gantt Chart / Project Plan

Task	Who	W3	W4	W5	W6	W7	Stu Vac	W8	W9	W10	W11	W12
Define requirements	Everyone											
Morphological table, pro/con analysis for each sub-function	Arabella and Teammate 1, Teammate 4 and Teammate 3 to finalise pro/con analysis											
Overall concept configurations	Everyone does at least 1 with different choices from each sub-function, 3 most feasible/ interesting chosen to present											
Scoring Matrix, decide on design (assignment 3)	Arabella and Teammate 1											
Initial rough drawings and calculations	Arabella											
CAD and technical drawings	Arabella											
Create physical catapult	Everyone											
Test/refine catapult	Everyone											
Milestones/deadlines												
Assign. 3 submission												
Assign. 4 submission												

Morphological Table

Sub-function	Solution Idea 1	Solution Idea 2	Solution Idea 3	Solution Idea 4	Solution Idea 5
1. Provide mechanical energy and accelerate payload					
Pros	Very simple, low cost, no moving parts	Simple, easy to adjust power by lifting the pendulum higher	Reliable and easily adjustable power and angle	Easy to use and make	Easy to use and predictable in launch pattern
Cons	Not adjustable in power or angle of launch	Can't adjust angle easily, hard to set pendulum height	Not safe as you can get your finger stuck between the coils	Hard to consistently use a certain power level, friction on walls	Complex design with multiple moving parts, elastic/rubber bands can decay and lose elasticity over time
2. Adjust throwing distance					
Pros	Can set at one angle and maintain angle reliably	Desired launch angle can be calculated quickly with projectile motion equations. Easy to set and measure.	Very Reliable and precise force given	Precise, Easy to make and simple	Catapult always retains one launch angle based upon length
Cons	Ball can lose energy due to a sudden angle change (bouncing)	Being able to move stopping bar may make it weaker, leading to potential energy losses or breakages when arm collides with it.	Time-consuming to re-measure force every time	May complicate release system of ball	New string needed to change angle, extra member that can fail

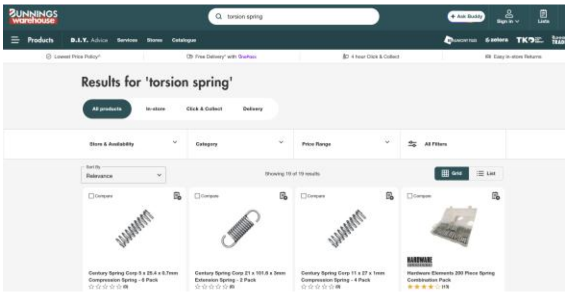
Sub-function	Solution Idea 1	Solution Idea 2	Solution Idea 3	Solution Idea 4	Solution Idea 5
3. Restrain (and launch) loaded catapult					
Pros	Easy to operate remotely with attached rope or long bar, reliably holds arm in place.	Configurable distance for restraint, can be release easily by pulling or cutting string, provides even force.	Easy to engage and disengage the latch	One simple part, can be remotely released with a rope or lever	Very cheap and easy to find in stores
Cons	Significant force required, due to friction, to retract bar. May lead to uneven trajectory with one side being released first.	Difficult to tie string quickly and reliably, quick-release knot may slip over time or thin string may fray.	Rubber shape would have to be made custom. Soft shape might warp and release unintentionally over time or with large forces.	Unbalanced, only restrains on one side, held almost entirely by friction of hinge	Depending on what you buy it could be overkill or underpowered for holding the tension. Hard to release arm reliably
4. Allow for easy dis-/assembly					
Pros	Allows for an easily foldable design that's very quick to assemble (arm attaches to base)	Super cheap and easy to make	Allows catapult to be disassembled into basic members for optimal space-use in container	Allows for members longer than maximum storage dimensions, folding design removes need for total disassembly	Very quick to assemble. Simple design minimises space catapult takes up in storage
Cons	Complex design that can jam and break	Arm only held by friction of the screw, needs to be un- and re-tightened every time the mechanism is folded	Slow to reassemble, friction-reliance may lead to catapult breaking apart	Many moving parts equate to structural weaknesses, may warp or	Any undesired movement such as jumping could make arm detach from base. Friction of weight resting on base.
5. Prevent catapult movement					
Pros	Simple, provides friction resistance against motion in all directions	Lowers centre of gravity to reduce tipping, increased mass reduces acceleration of catapult body	Counters forwards motion of catapult, directly opposes forces	Very solid base as it prevents horizontal movement well	Not only lowers horizontal movement but also reduces stress fatigue on materials by reducing shocks
Cons	Of limited use in preventing tipping, may not be effective against large forces	Makes catapult less practical to move, may affect assembly and storage	Increases required force due to added mass, increases the rotation of the system	Can damage the floor, safety hazard due to cutting yourself and expensive	Hard to implement into design

Scoring Matrix

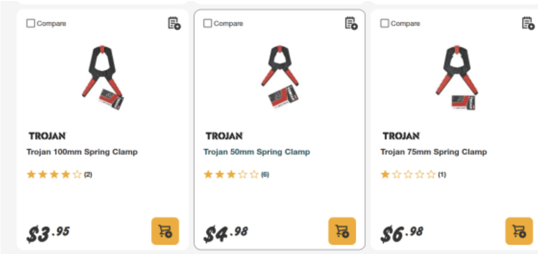
Selection Criteria	Weight	Configuration Assessment					
		Config 1		Config 2		Config 3	
		Rating	Score	Rating	Score	Rating	Score
Maximum throwing distance	5	4	20	4	20	4	20
Adjustability of throwing distance	5	5	25	2	10	4	20
Ease of assembly	4	4	16	5	20	4	16
Ease of manufacturing	3	3	12	5	20	3	12
Safety of use	4	4	16	5	20	3	12
		Total	89	Total	90	Total	80

Finding Torsion Springs

Bunnings (and other hardware stores such as Mitre 10) do not sell any plain torsion springs

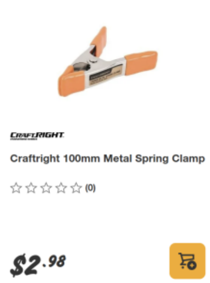


They do sell them in hinges and clamps such as these, which might be a good option.



Dedicated spring manufacturers could make us custom springs which exactly fit our requirements, but this is too expensive and time-consuming for this project, and we only really need 1 spring.

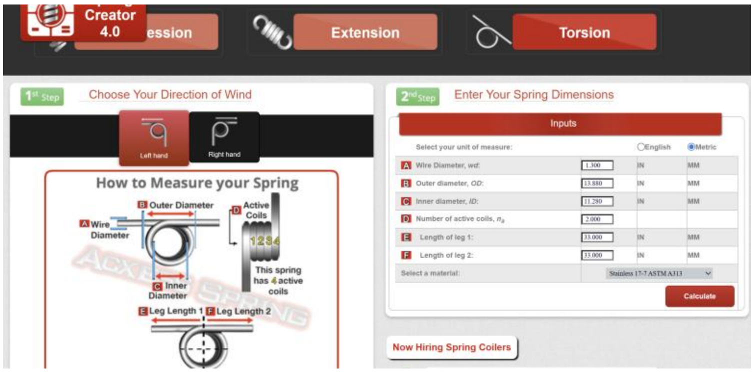
The Craftright 100mm spring clamp was purchased from Bunnings, then disassembled to remove the torsion spring in it. However, the internal diameter of this spring was too small, fitting on hinges with a maximum of 5mm diameter only, and it was not deemed appropriate for the project.



Fortunately, better sized springs were salvaged from an old IKEA clothes dryer, with a slightly lower spring coefficient but able to fit around 10mm rods for a sturdier catapult design.

Torsion spring constant estimate

Estimating spring constant using an online calculator



Rates & Torques

Rate per degree :	5.939 N-mm/Degree
Spring Rate (or Spring constant) per 360 degrees, $k_{360\ degrees}$:	2,137.864 N-mm/360 Degrees
Maximum torque possible, $Torque_{max}$:	268.732 N-mm

Physical Dimensions

Diameter of spring wire, d:	1.300 mm
Outer diameter of spring, D_{outer} :	13.880 mm
Inner diameter of spring, D_{inner} :	11.280 mm
Mean diameter of spring, D_{mean} :	12.580 mm
Number of active coils, n_a :	2.000
Body length, L_{body} :	3.900 mm
Length of leg 1:	33.000 mm
Length of leg 2:	33.000 mm
Total leg length:	66.000 mm
Direction of wind :	Left hand
Spring index, C :	9.677

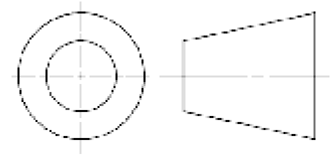
k=5.939 N.mm/degree = 5.939x10^-3 N.m/degree = 0.34 N.m/radian

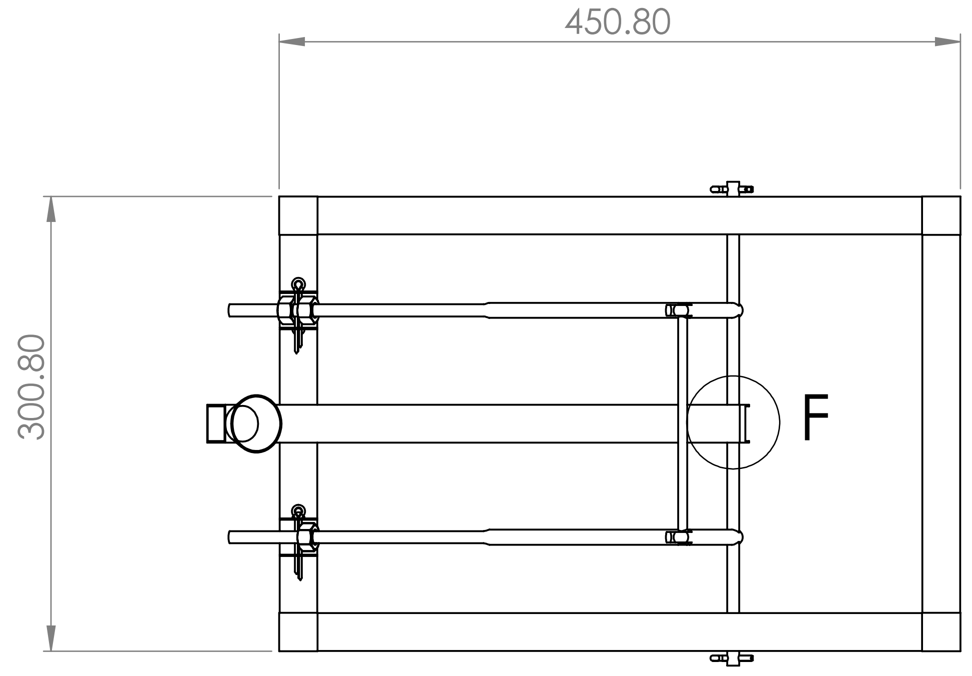
Appropriate shear forces

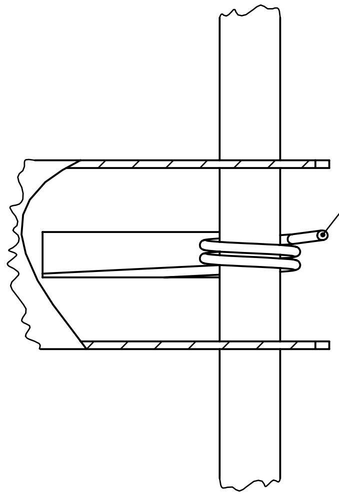
To determine whether the shear force exerted on the hinge pin in the adjustable stopping-bar mechanism is appropriate, it is necessary to compare the calculated shear stress with known material strengths. This will allow us to verify that there will be no warping or breaking of the pin, and that the design can be used reliably as per safety requirements.

It is assumed, realistically, that the hinge pin will be made out of stainless steel, but for this calculation we will suppose it to be a weaker metal, to introduce a factor of safety. According to UniPunch (2026), the shear strength of steel is between 276 and 1380 MPa, while the lowest reported shear strength, for a 1100-0 aluminium alloy, was 62 MPa. As such, if the shear stress exerted on the hinge pin exceeds 62 MPa, the component will have to be redesigned. Otherwise, it is acceptable.

9 Appendix B: Technical Drawings



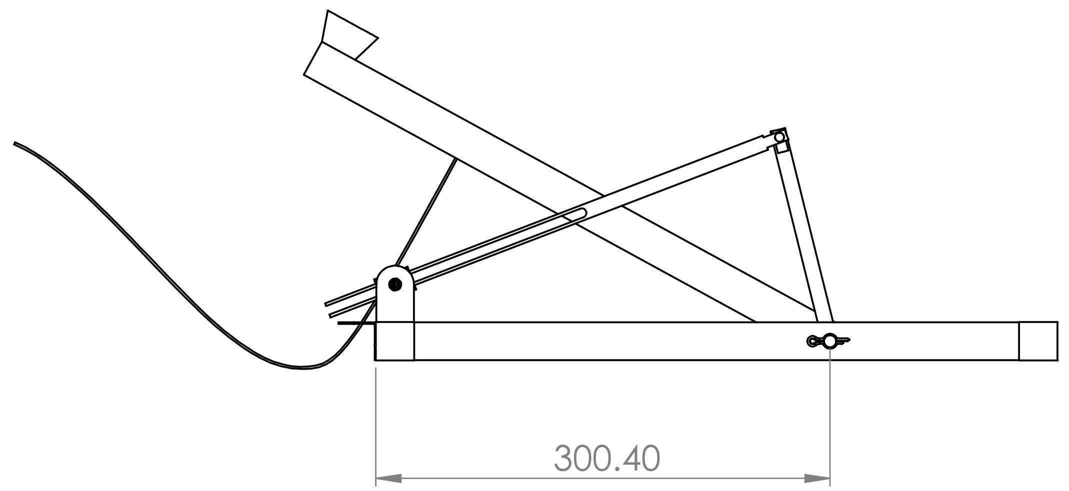


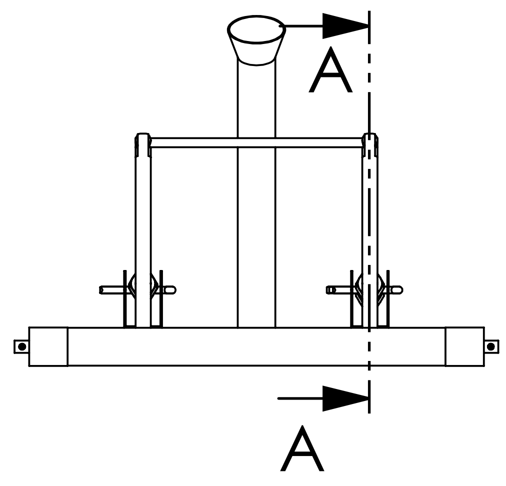


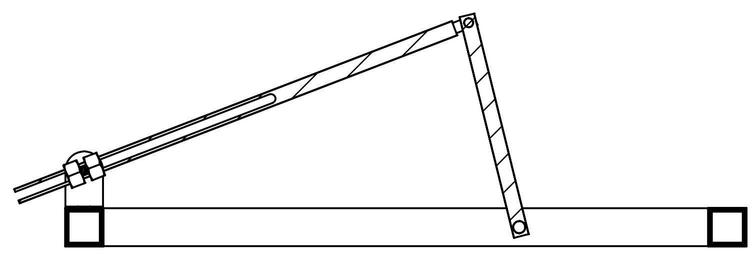
Note: Torsion spring sits inside catapult arm, threaded onto central hinge. One end presses on upper edge of inside of arm, other presses against the floor through a slot in the arm.

DETAIL F

SCALE 1 : 1

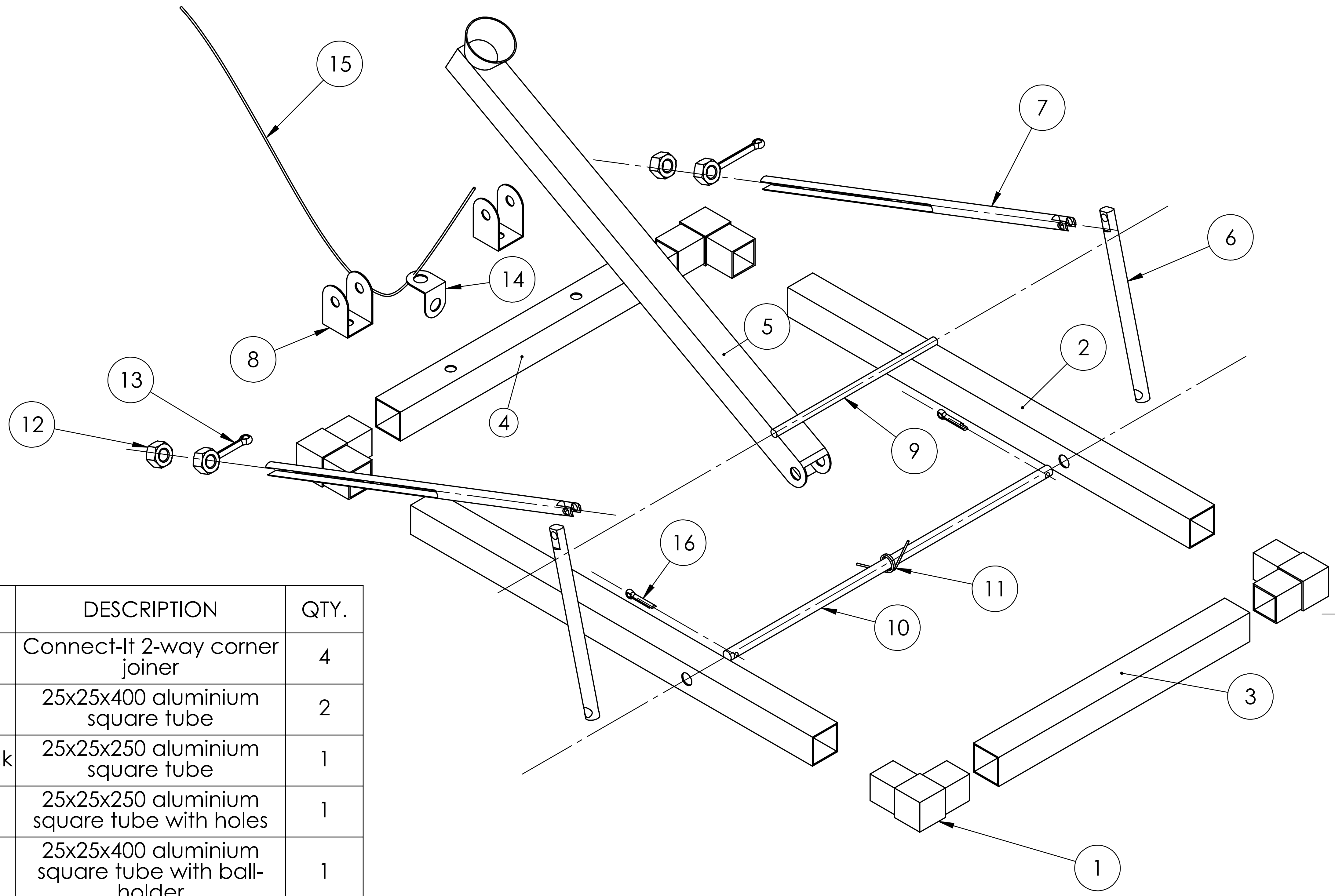






SECTION A-A

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH:		DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE	TITLE: catapult frame assembly	
DRAWN	Arabella Simpson		14/05/2026		
CHK'D					
APPV'D					
MFG					
Q.A			MATERIAL:	DWG NO.	A3
			WEIGHT:	SCALE:1:5	SHEET 1 OF 14



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	connectors	Connect-It 2-way corner joiner	4
2	long_span_catapult_base	25x25x400 aluminium square tube	2
3	short_span_catapult_base_back	25x25x250 aluminium square tube	1
4	short_span_catapult_base	25x25x250 aluminium square tube with holes	1
5	catapult_arm	25x25x400 aluminium square tube with ball-holder	1
6	stopping_bar_vertical	∅10x150 rod	2
7	stopping_bar_adjustable_rod	∅10x325 threaded rod	2
8	hinge_bracket	U-shape bracket	2
9	stopping_bar	∅6x160 rod	1
10	central_hinge	∅5x304 rod	1
11	torsion_spring		1
12	AS 1112.3 C- M10-N	M10 hexagonal nut	4
13	AS 1236 PSM - 5 x 36	∅5x36 split pin	2
14	string bracket	L-shape bracket	1
15	string	3m nylon string	1
16	AS 1236 PSM - 4 x 16	∅4x16 split pin	2

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN	Arabella Simpson		14/05/2026
CHK'D			
APPV'D			
MFG			
Q.A			

MATERIAL:

WEIGHT:

TITLE:

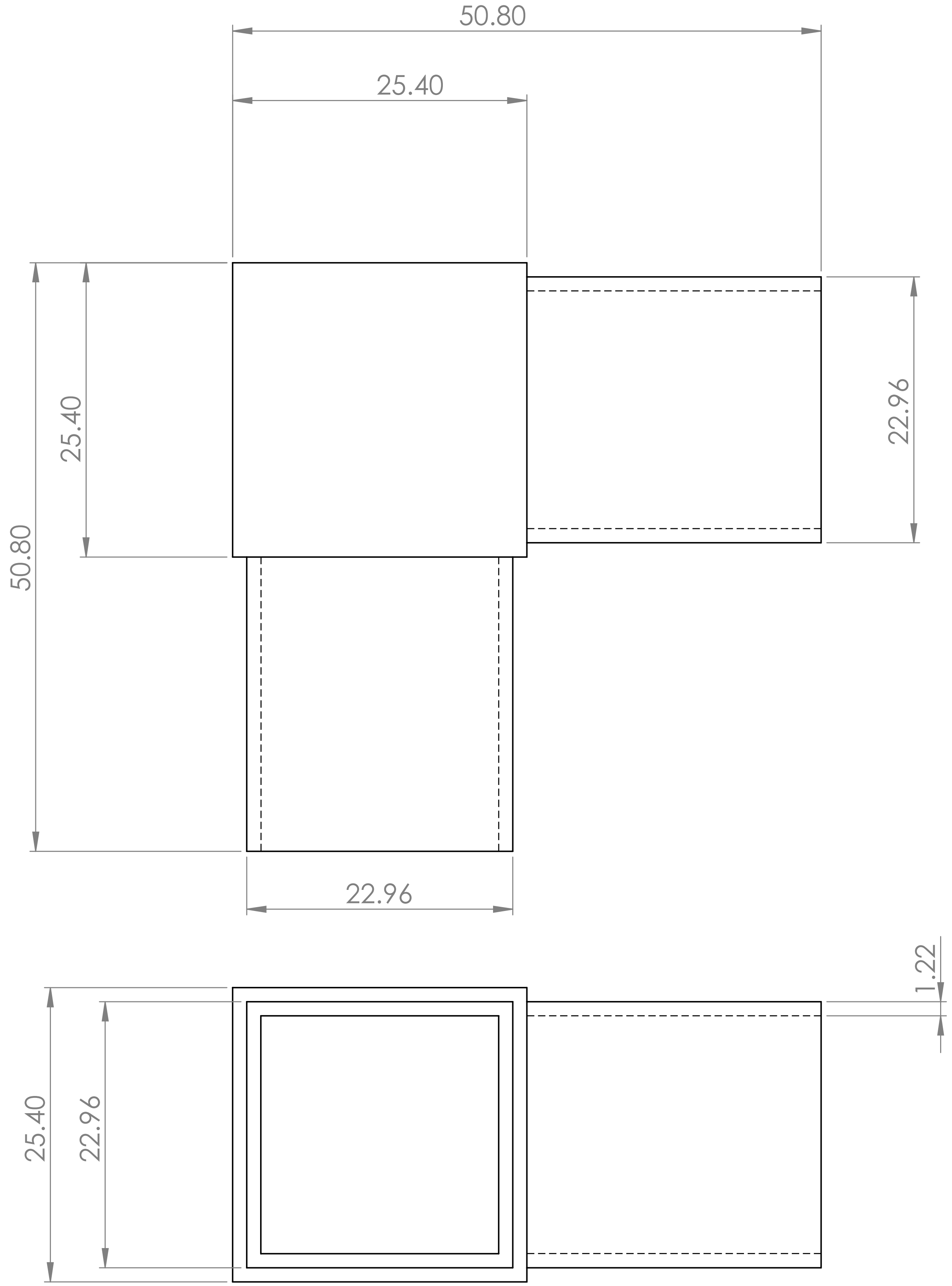
catapult frame assembly: exploded isometric view

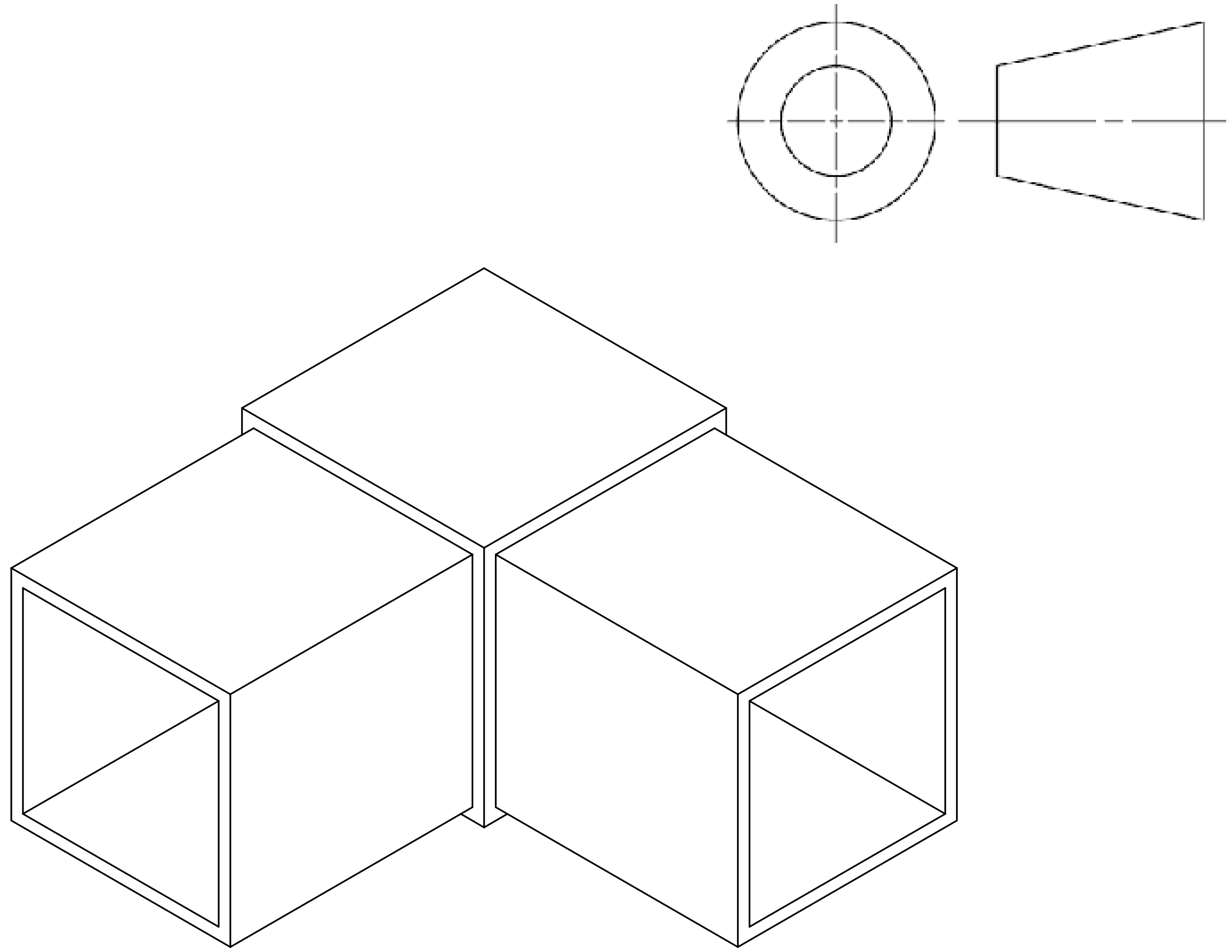
DWG NO.

A3

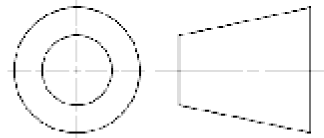
SCALE:1:10

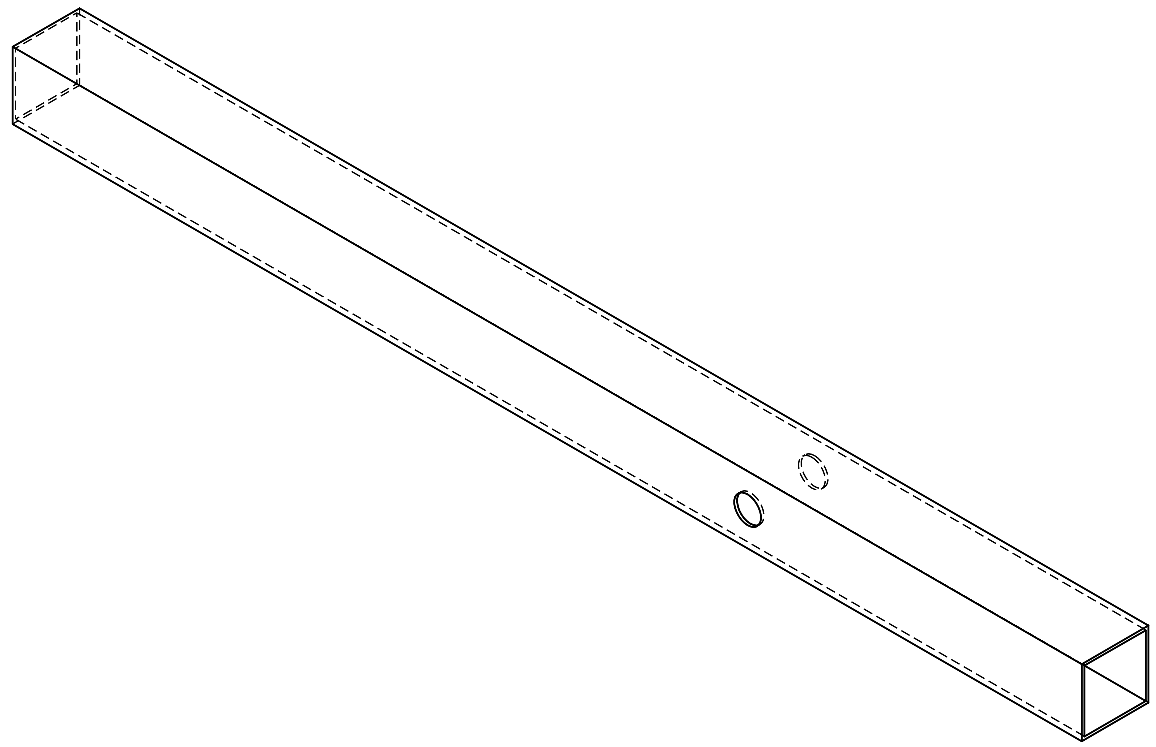
SHEET 2 OF 14

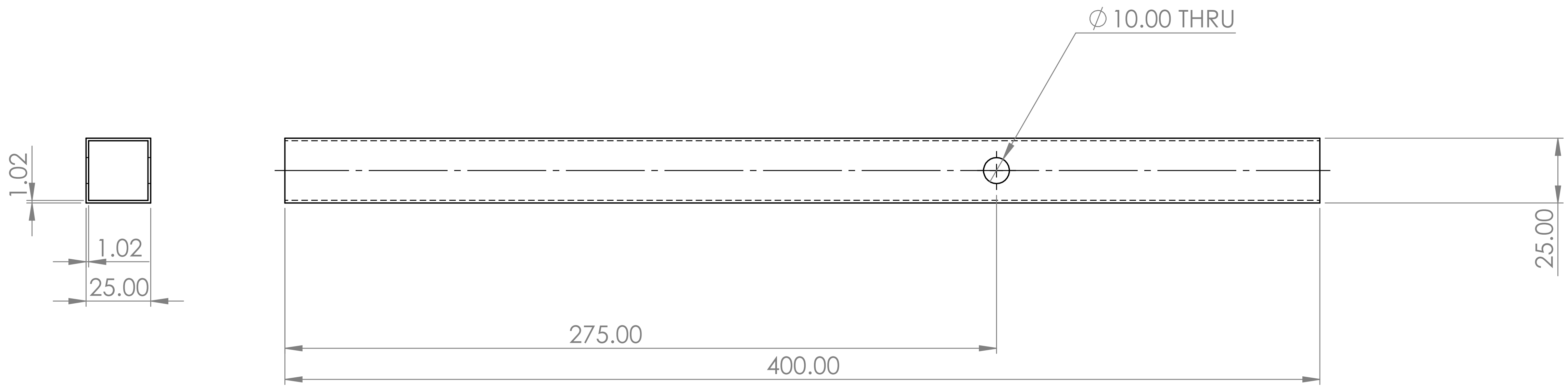




UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH:			DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE		TITLE:	
DRAWN	Arabella Simpson		12/05/2026		Connect-It 2-way corner joiner	
CHK'D						
APPV'D						
MFG						
Q.A				MATERIAL:	DWG NO.	A3
				WEIGHT:	SCALE:2:1	SHEET 3 OF 14







UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN	Arabella Simpson		11/05/2026
CHK'D			
APPV'D			
MFG			
Q.A			

MATERIAL:

WEIGHT:

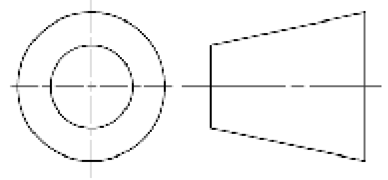
TITLE:
long_span_catapult_base: 400mm aluminium 25x25 tubing

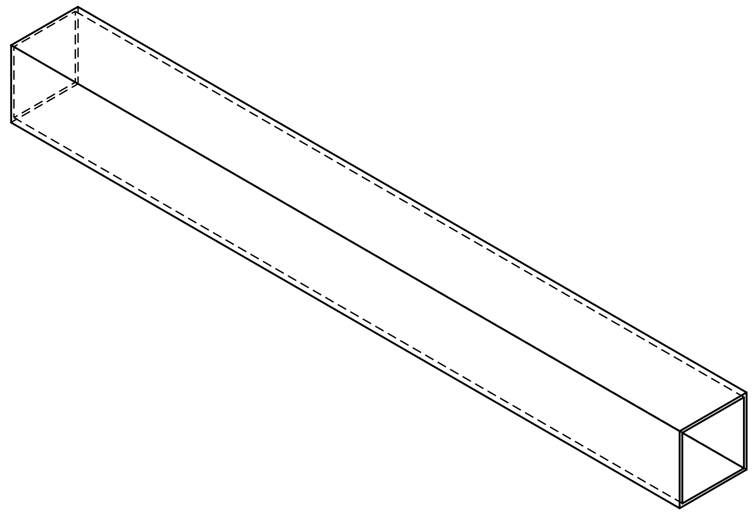
DWG NO.

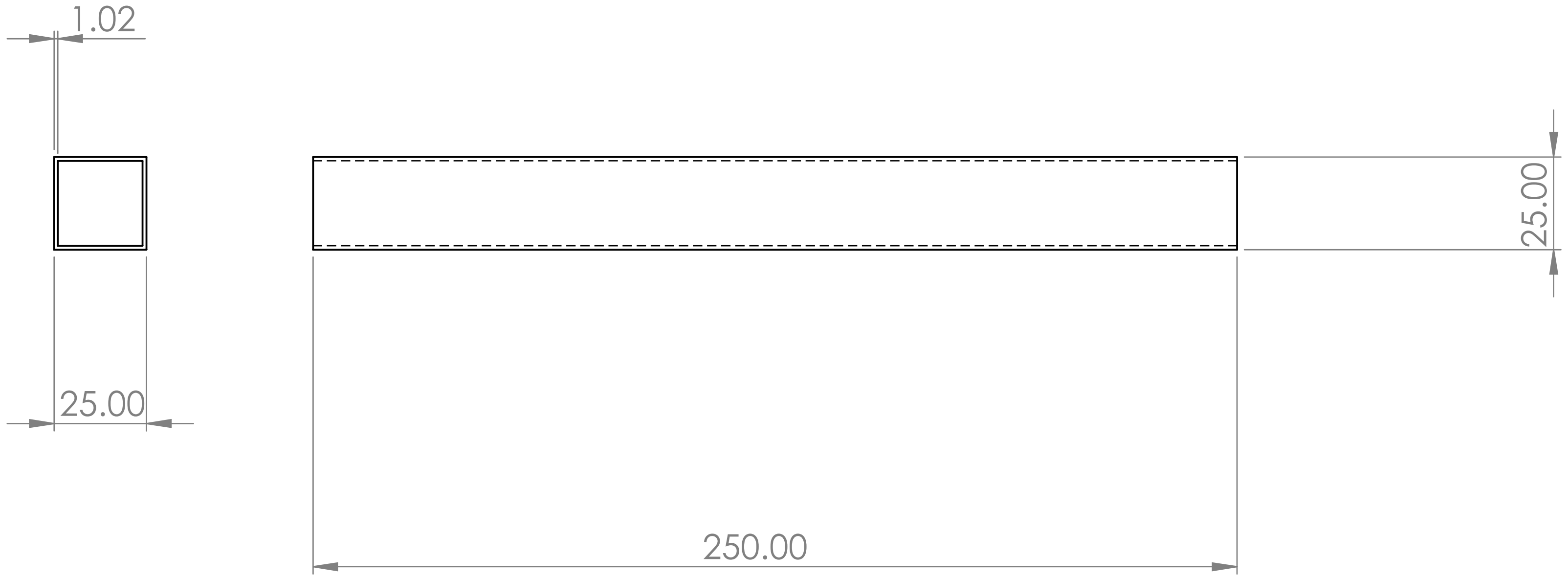
A3

SCALE:1:5

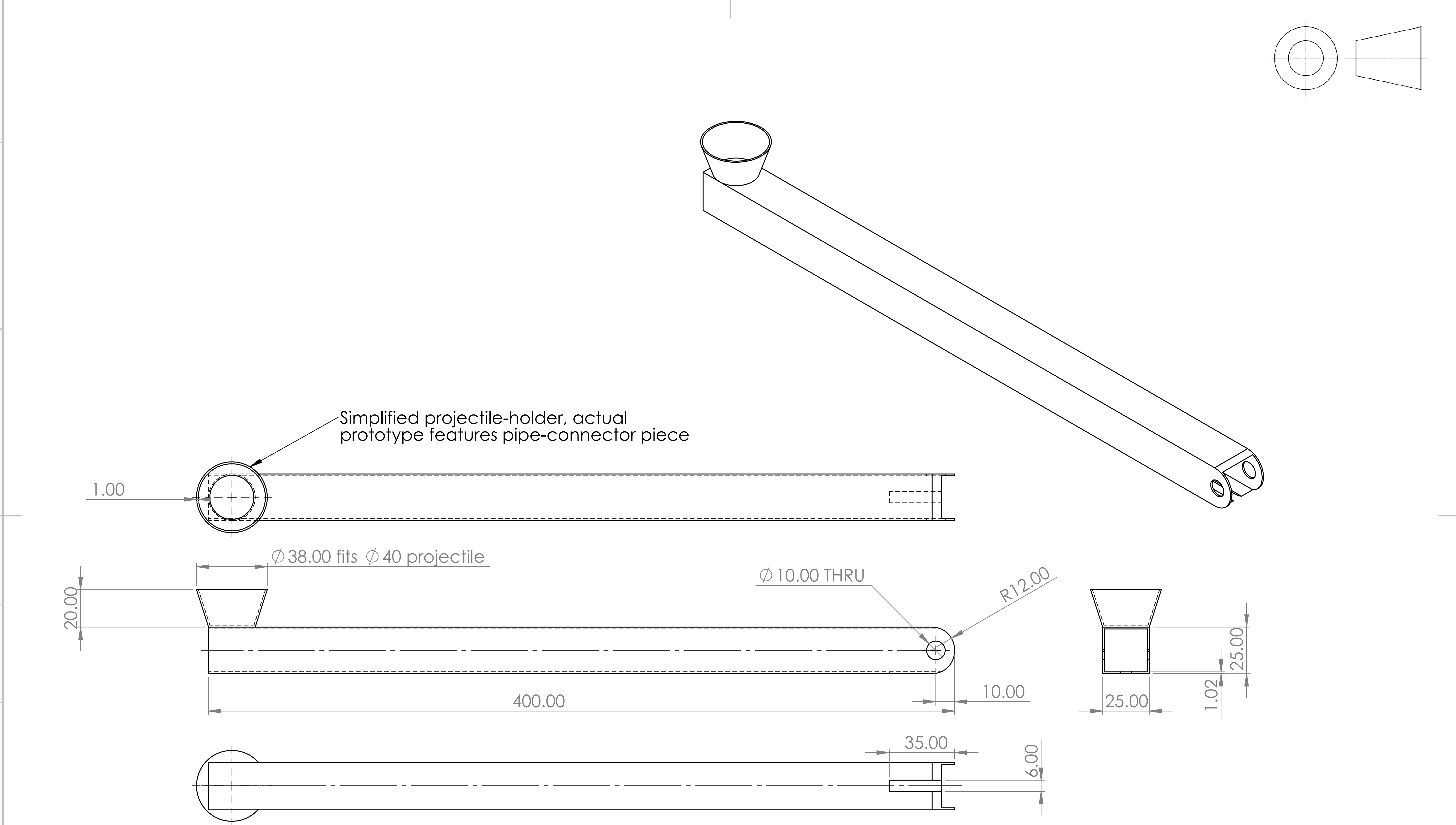
SHEET 4 OF 14



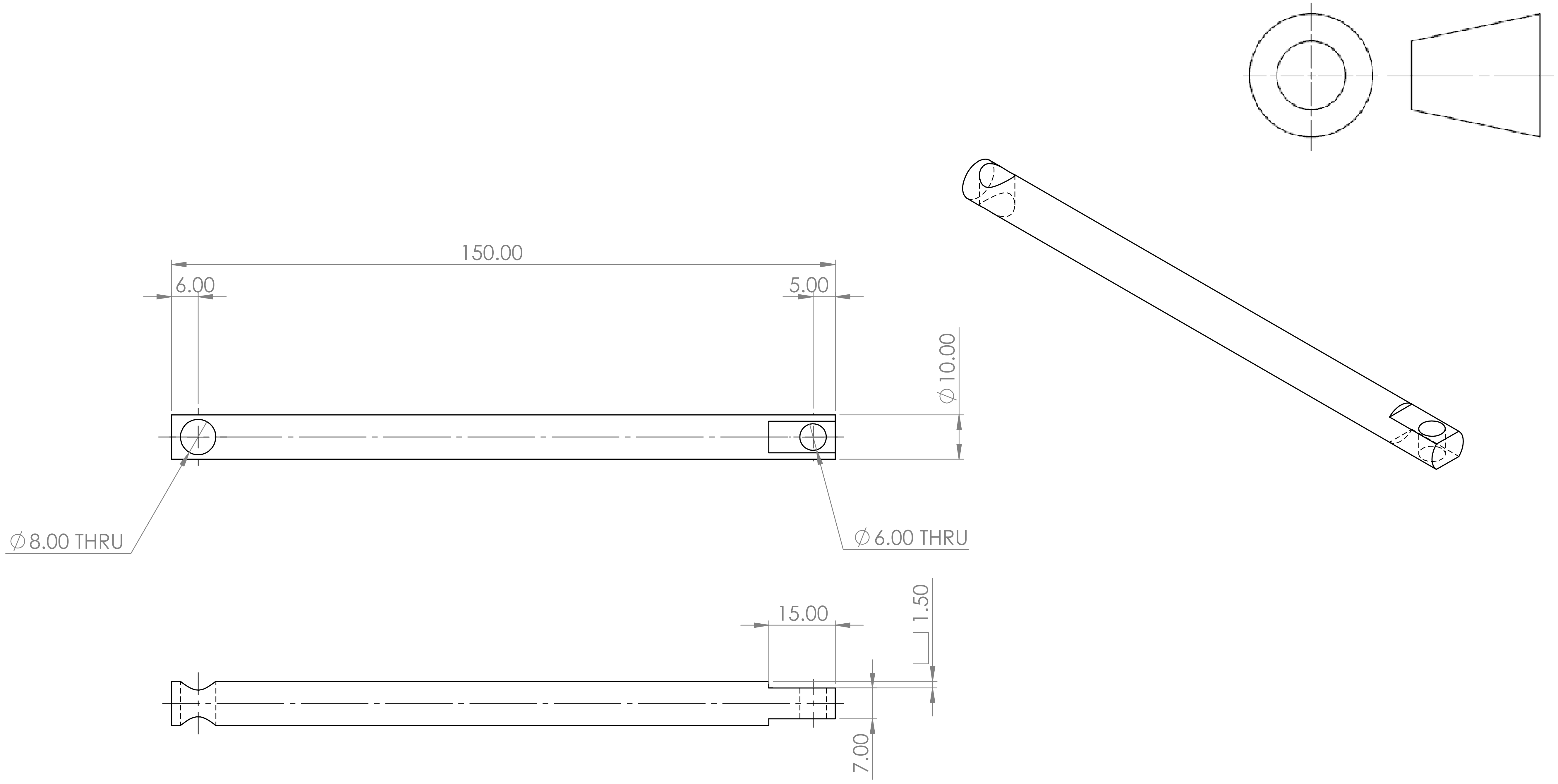




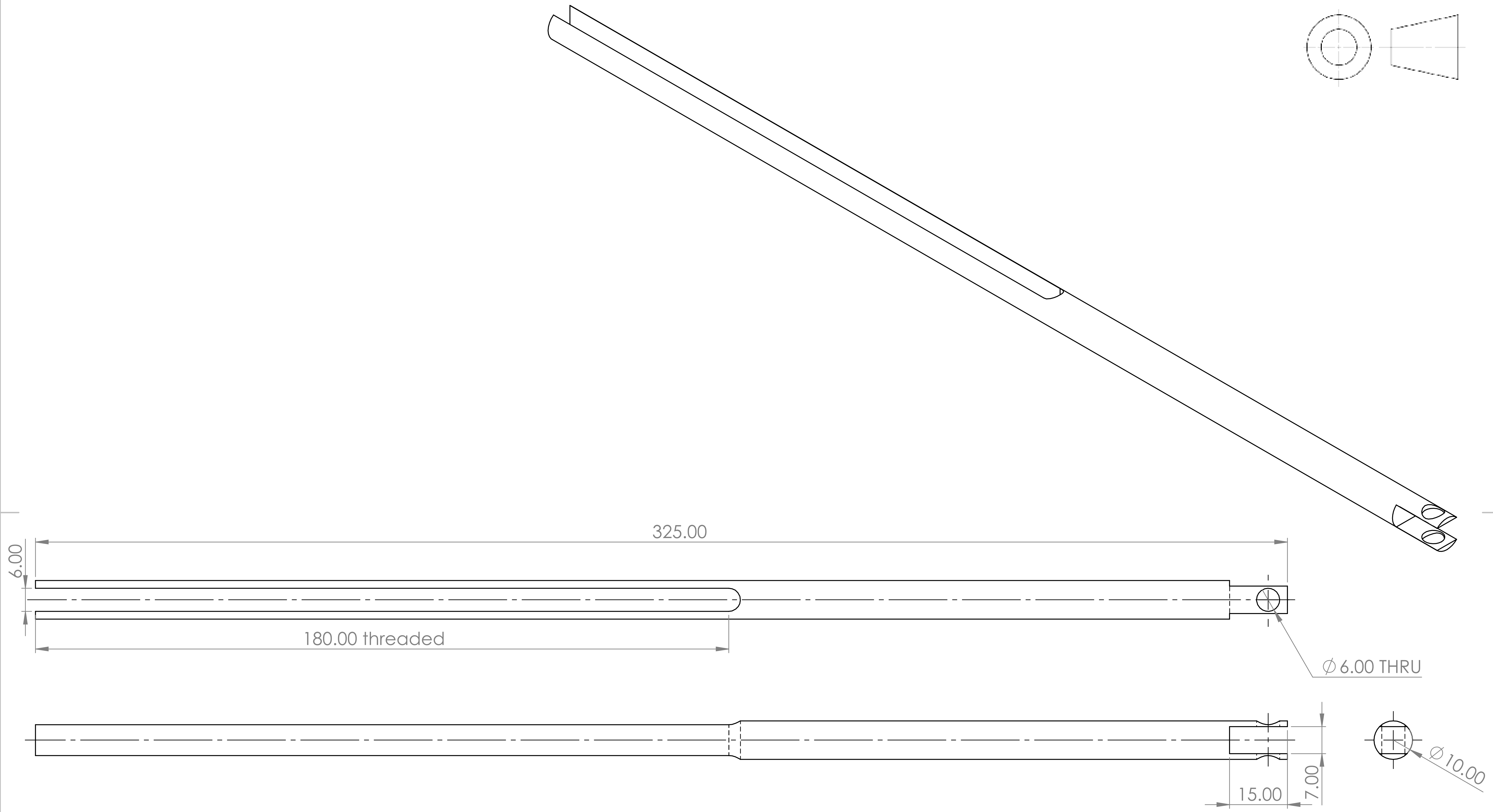
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH:	DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION	
	NAME	SIGNATURE	DATE		TITLE:
DRAWN	Arabella Simpson		12/05/2026		short_span_catapult_base_back: 25x25x250mm aluminium tubing
CHK'D					
APPV'D					
MFG					
Q.A				MATERIAL:	DWG NO. A3
				WEIGHT:	SCALE:1:2 SHEET 5 OF 14



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH:		DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE	TITLE:	
DRAWN	Arabella Simpson		12/05/2026	catapult_arm: 25x25x400 aluminium tubing	
CHK'D					
APPV'D					
MFG					
Q.A				MATERIAL:	DWG NO. A3
				WEIGHT:	SCALE:1:5 SHEET 6 OF 14



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH:		DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE	TITLE:	
DRAWN	Arabella Simpson		12/05/2026	stopping_bar_vertical: ∅ 10 x 150 aluminium rod	
CHK'D					
APPV'D					
MFG					
Q.A			MATERIAL:	DWG NO.	A3
			WEIGHT:	SCALE:1:1	SHEET 7 OF 14



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN	Arabella Simpson		12/05/2026
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

stopping_bar_adjustable_rod:
∅ 10x325 partially threaded rod

MATERIAL:

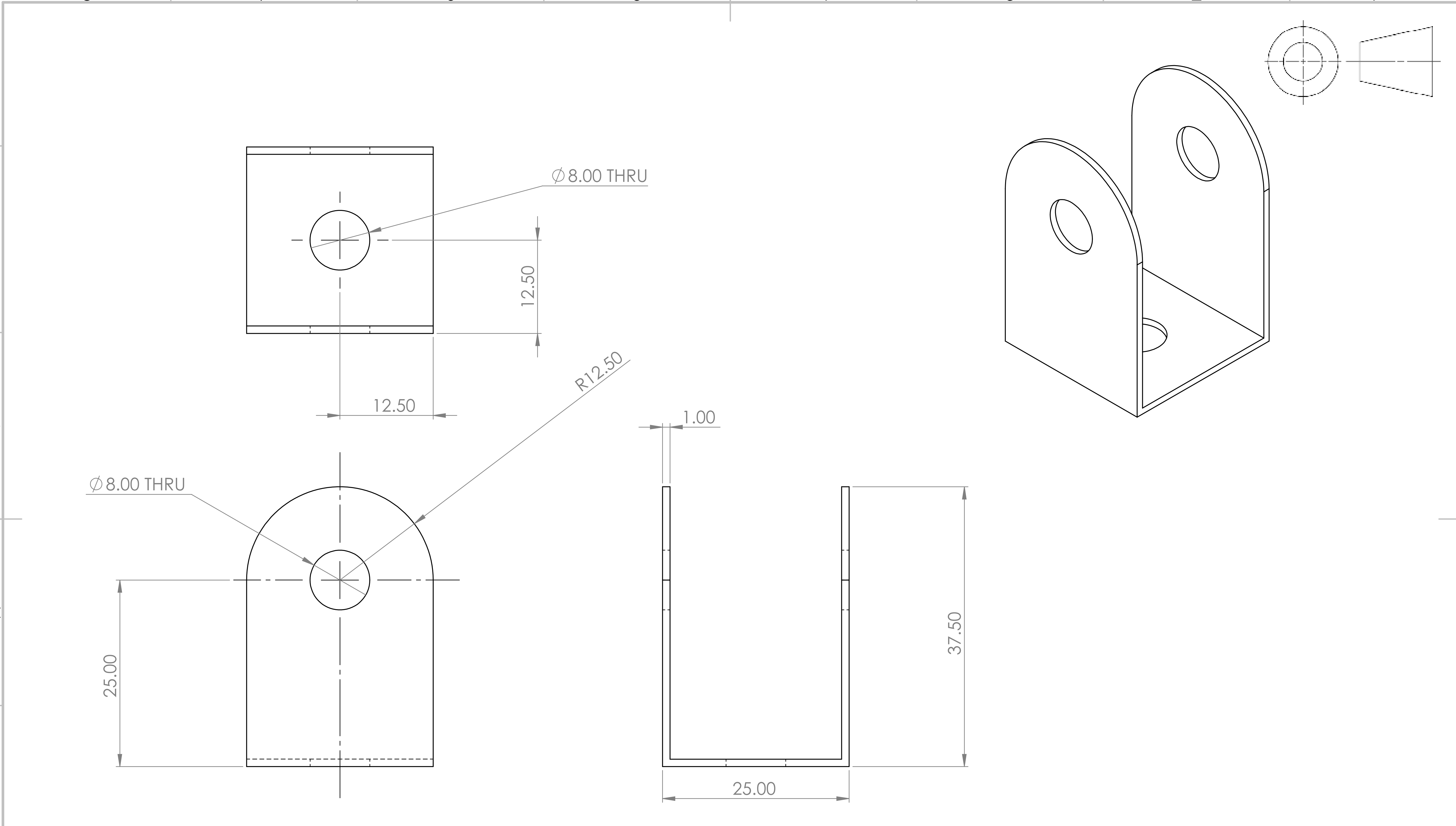
WEIGHT:

DWG NO.

A3

SCALE: 1:1

SHEET 8 OF 14



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN	Arabella Simpson		12/05/2026
CHK'D			
APPV'D			
MFG			
Q.A			

MATERIAL:

WEIGHT:

TITLE:

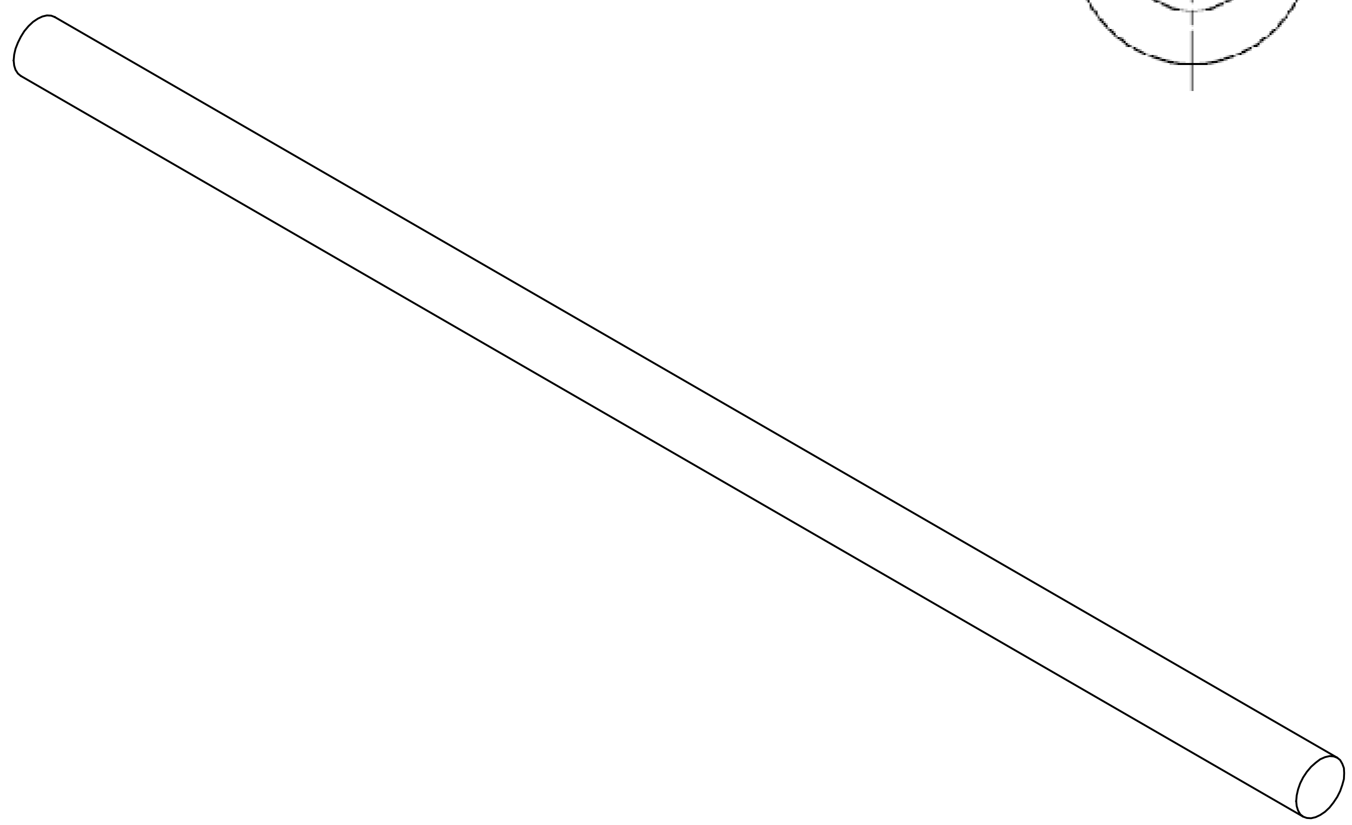
hinge_bracket

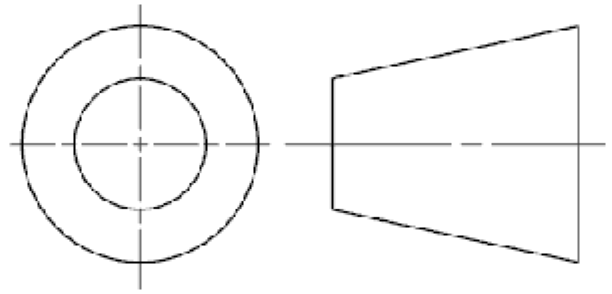
DWG NO.

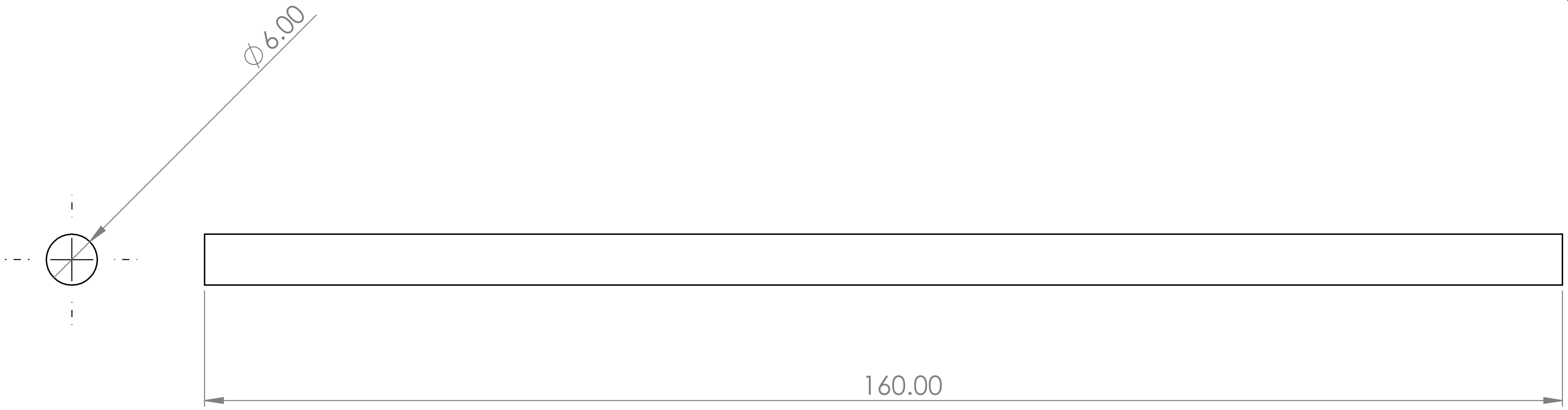
A3

SCALE:2:1

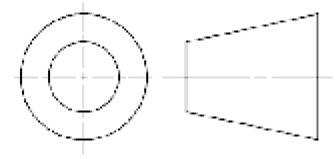
SHEET 9 OF 14

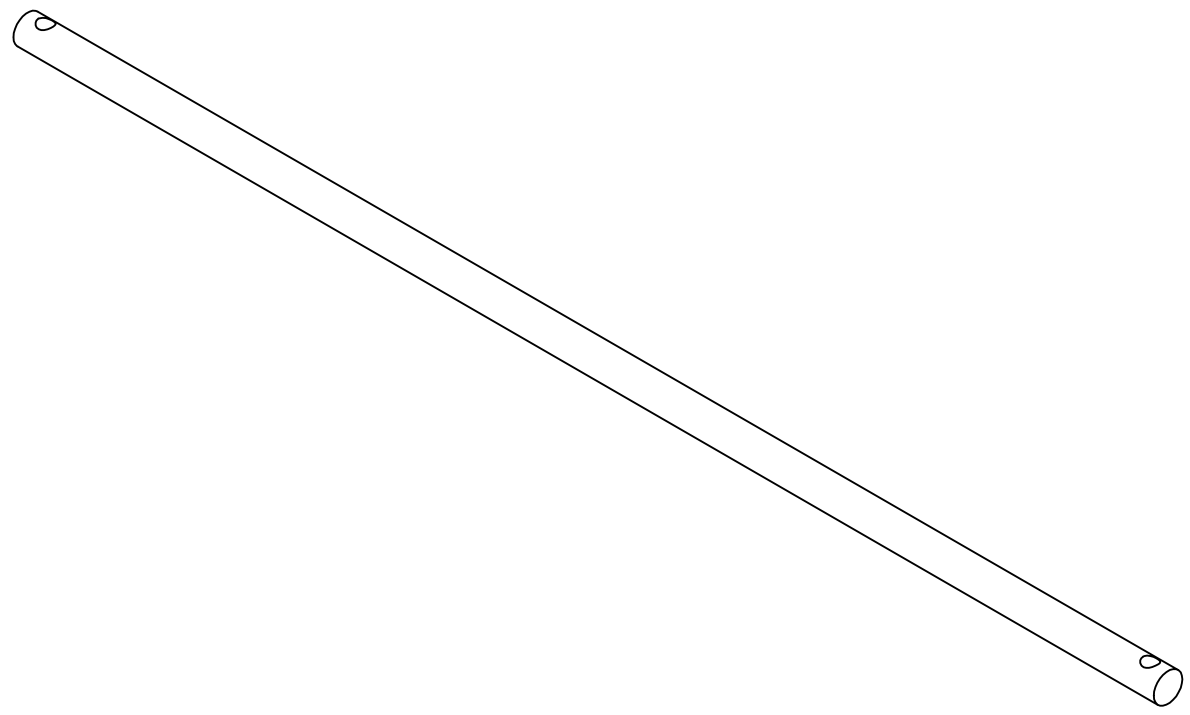


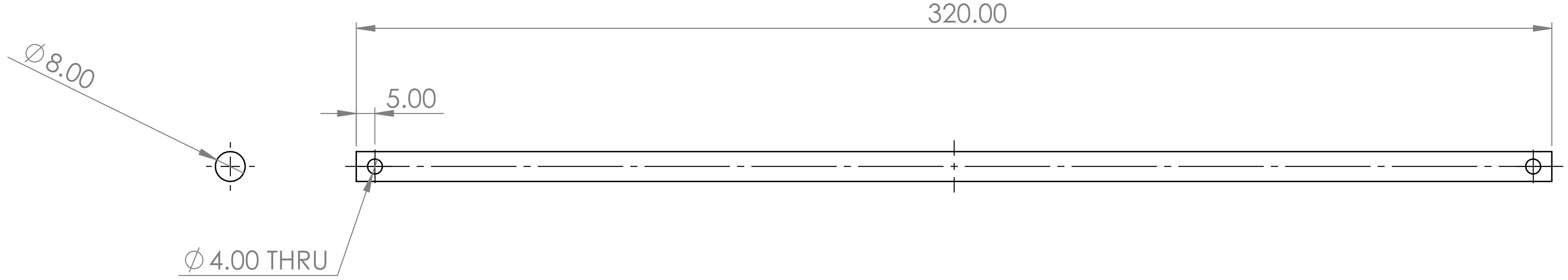




UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH:		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE		TITLE: stopping_bar: ∅ 6x160 rod	
DRAWN	Arabella Simpson		12/05/2026			
CHK'D						
APPV'D						
MFG						
Q.A				MATERIAL:	DWG NO.	A3
				WEIGHT:	SCALE:1:2	SHEET 10 OF 14







UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN	Arabella Simpson		12/05/2026
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

central_hinge: ⌀ 8x320 rod

MATERIAL:

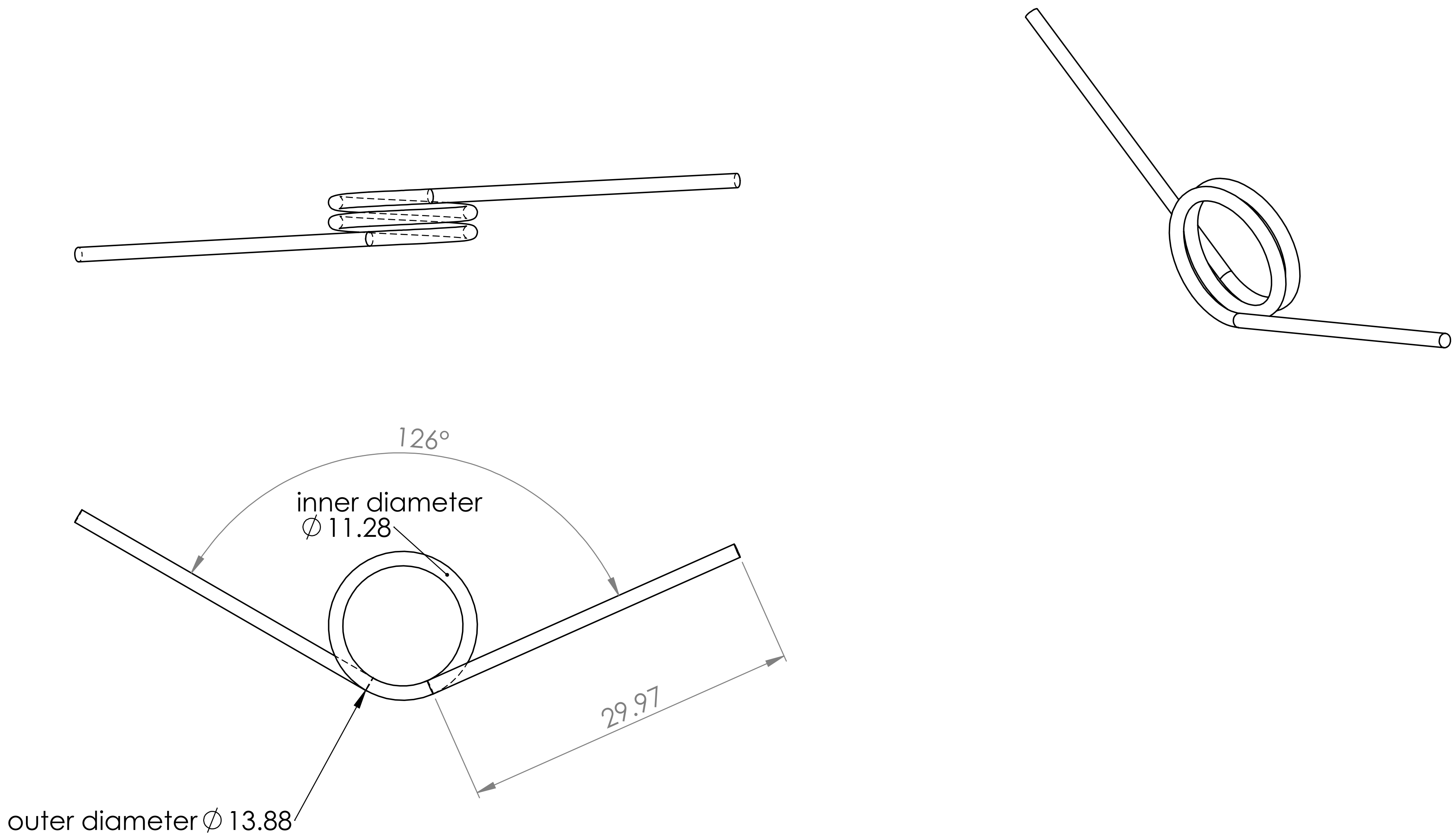
DWG NO.

A3

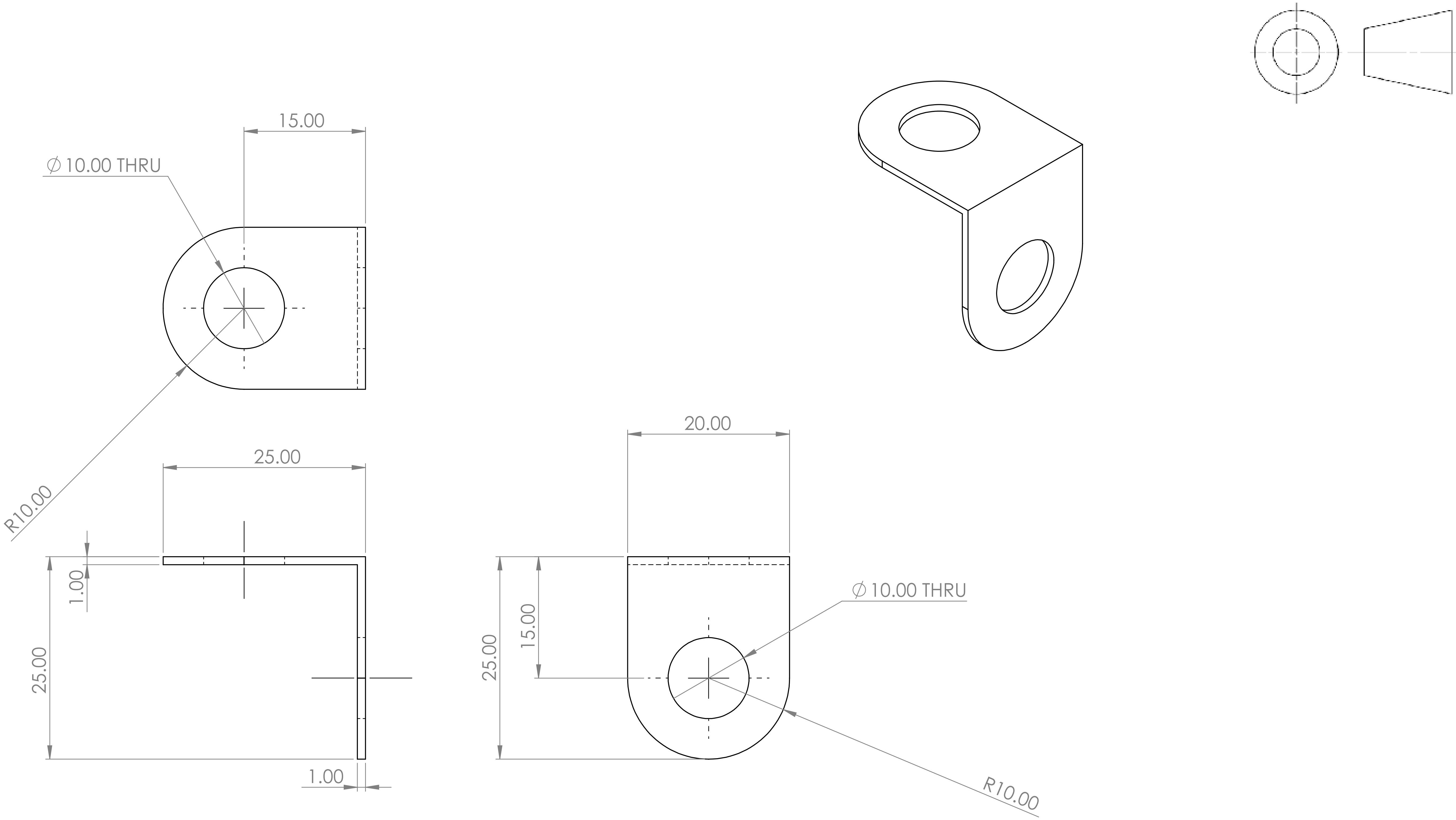
WEIGHT:

SCALE: 2:3

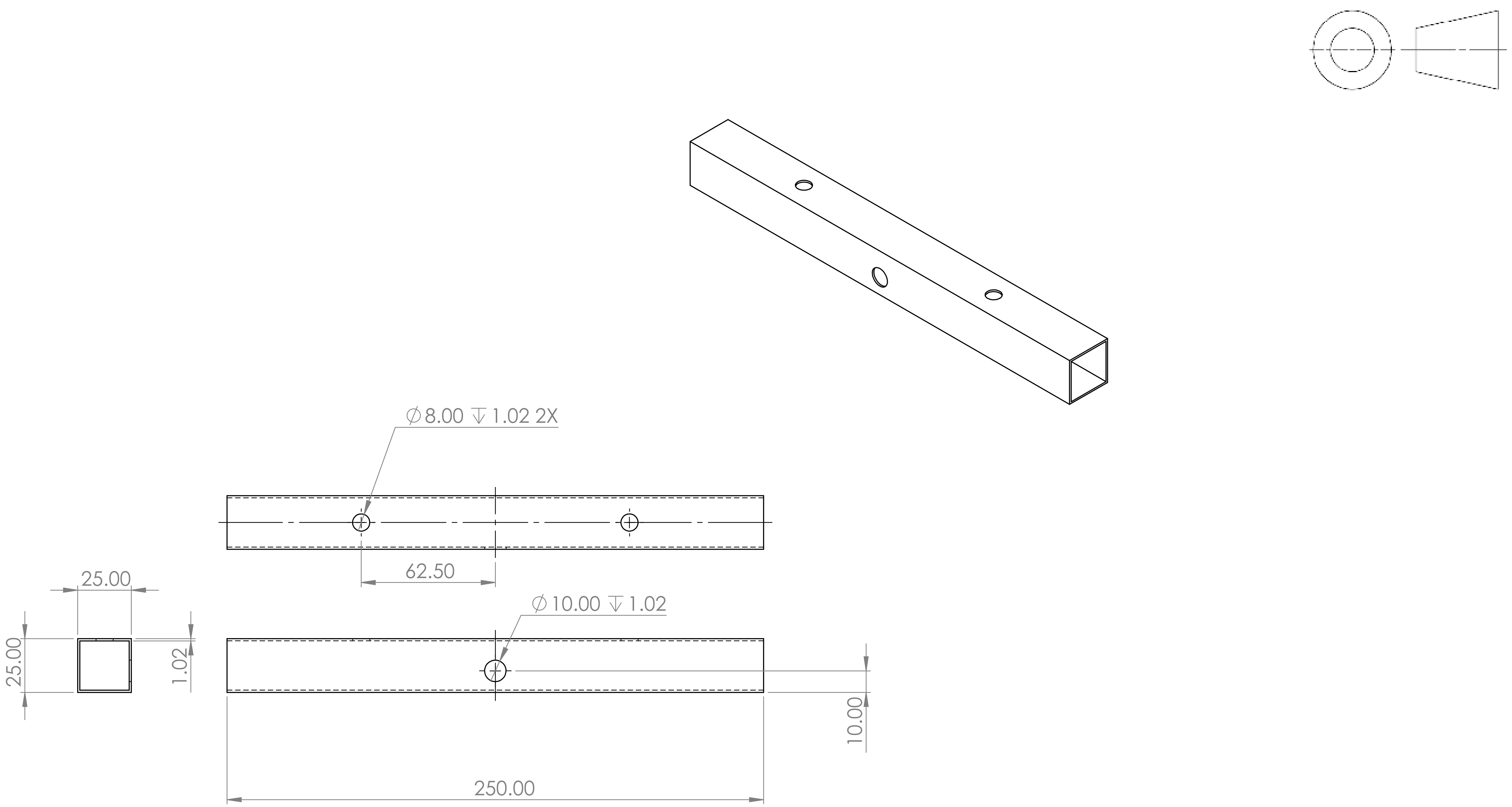
SHEET 11 OF 14



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH:		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE		TITLE:	
DRAWN	Arabella Simpson		12/05/2026		torsion_spring	
CHK'D						
APPV'D						
MFG						
Q.A				MATERIAL:	DWG NO.	A3
				WEIGHT:	SCALE:2:1	SHEET 12 OF 14



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH:		DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE	TITLE:	
DRAWN	Arabella Simpson		12/05/2026	string bracket: L-shaped bracket	
CHK'D					
APPV'D					
MFG					
Q.A			MATERIAL:	DWG NO.	A3
			WEIGHT:	SCALE:2:1	SHEET 13 OF 14



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN	Arabella Simpson		12/05/2026
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

short_span_catapult_base: 25x25x250 aluminium tubing

MATERIAL:

DWG NO.

A3

WEIGHT:

SCALE:1:2

SHEET 14 OF 14